BANGLADESH ARMY UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAUST)



SYLLABUS

BACHELOR OF SCIENCE IN
COMPUTER SCIENCE AND ENGINEERING (CSE)

REVISED ON AUGUST2015

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)
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CONTENTS

CHAPTER 1
GENERAL INFORMATION

1.1 Introduction …… 1
1.2 Aim …… 1
1.3 Objectives …… 1
1.4 Location …… 1
1.5 Eligibility of Students for Admission in BAUST …… 2
1.6 Admission Procedure …… 2
1.6.1 Syllabus for Admission Test …… 2
1.6.2 Final Selection …… 2
1.6.3 Medical Checkup …… 3

CHAPTER 2
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2.1 Introduction …… 4
2.2 Laboratory Facility of the Department …… 4

CHAPTER 3
RULES AND REGULATIONS FOR UNDERGRADUATE PROGRAMME

3.1 Number of Terms in a Year(Level) …… 5
3.2 Duration of Terms …… 5
3.3 Course Pattern and Credit Structure …… 5
3.3.1 Course Designation System …… 5
3.3.2 Assignment of Credits …… 6
3.3.3 Type of Courses …… 6
3.4 The Grading System …… 7
3.4.1 The Letter Grade …… 7
3.4.2 Distribution of Marks …… 8
3.4.3 Calculation of CGPA …… 8
3.4.4 Minimum Earned Credit and GPA Requirement for Obtaining Degree…… 9
3.5 Attendance …… 9
3.6 Teacher-Student Interaction …… 10
3.7 Conduct and Discipline …… 10

CHAPTER 4
COURSE REQUIREMENTS FOR UNDERGRADUATE COMPUTER SCIENCE AND ENGINEERING STUDENTS

4.1 Introduction …… 11
4.2 Core Courses …… 11
4.2.1 List of Core Courses-CSE …… 11

4.2.2 List of Core Courses-A&S 13
4.2.3 List of Core Courses-EEE 13
4.2.4 List of Core Courses-ME 13

LEVEL-1 TERM-1 14
LEVEL-1 TERM-2 14
LEVEL-2 TERM-1 15
LEVEL-2 TERM-2 15
LEVEL-3 TERM-1 16
LEVEL-3 TERM-2 16
LEVEL-4 TERM-1 17
LEVEL-4 TERM-2 17
OPTION-I 17
OPTION-II 18
OPTION-III 18
SUMMARY 19

CHAPTER 5
DETAIL OUTLINE OF UNDERGRADUATE COURSES OFFERED BY DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LEVEL-1 TERM-1 20
LEVEL-1 TERM-2 24
LEVEL-2 TERM-1 27
LEVEL-2 TERM-2 30
LEVEL-3 TERM-1 33
LEVEL-3 TERM-2 36
LEVEL-4 TERM-1 40
LEVEL-4 TERM-2 42
OPTION-I 44
OPTION-II 49
OPTION-III 53

CHAPTER - 1

GENERAL INFORMATION

1.1 Introduction

With a view to meeting the increasing demand for the development and dissemination of engineering and technological know-how, Bangladesh Armed Forces established the Bangladesh Army University of Science and Technology (BAUST), Saidpur that promises to provide facilities for higher technical education for the students home and abroad. BAUST starts its journey on 15 February 2015 by offering four-year bachelor's degrees on Computer Science and Engineering (CSE), Electrical and Electronic Engineering (EEE), and Mechanical Engineering (ME).

1.2 Aim

The aim of BAUST is to conduct undergraduate courses in various disciplines of Engineering according to syllabi leading to Bachelor of Science in Engineering (B. Sc. Engineering) for the students from home and abroad.

1.3 Objectives

The objectives of BAUST are:

1.3.1 To offer the following courses with a view to meeting the increasing demands in the country:

(i) Four-years bachelor course in Computer Science and Engineering (CSE)
(ii) Four-years bachelor course in Electrical and Electronic Engineering (EEE) and
(iii) Four-years bachelor course in Mechanical Engineering (ME).

1.3.2 To produce skilled, well disciplined, self-motivated and dedicated engineers and computer professionals.

1.3.3 To make provisions for research and development and dissemination of knowledge in appropriate fields of science and technology.

1.4 Location

BAUST is located at Saidpur Cantonment, Saidpur, Nilphamari, a hub of knowledge for Bangladesh Armed Forces. Saidpur Cantonment is a calm and quiet education village and free from all possible pollution of a city life. Whistling birds on the tree branches and overall bounty of nature adds to the existing splendid academic atmosphere.

1.5 Eligibility of Students for Admission in BAUST

The students must fulfill the following requirements:

1.5.1 Applicants must have passed SSC and HSC (or equivalent) examination in Science group with minimum GPA of 3.5 in both but total of minimum 7.5.

1.5.2 Candidates who have passed HSC or equivalent exam in the year or one year before the notification for admission can apply. Candidates with more than one year break of study will not be eligible to apply.

1.5.3 For O Level/Junior Cambridge and A Level/Senior Cambridge background students, the applicants must have to qualify minimum 5 subjects in O Level/Junior Cambridge and 3 subjects including Mathematics, Physics and Chemistry in A Level/Senior Cambridge with minimum C grade in all subjects.

1.6 Admission Procedure

1.6.1 Syllabus for Admission Test

Admission test will be conducted on the basis of the syllabus of Mathematics, Physics, Chemistry and English (comprehension and functional) subjects of HSC examinations of all boards of secondary and higher secondary school certificates. Admission test will be conducted out of 100 marks and the distribution of marks is as follows:

Ser.	Subject	Marks
1	Mathematics	40%
2	Physics	30%
3	Chemistry	20%
4	English	10%
	Total:	**100%**

1.6.2 Final Selection

Minimum qualifying marks in the written admission test is 40%. Merit list of candidates for final selection and admission to university will be prepared on the basis of the following:

Written Admission Test	60%.
GPA of SSC/ Dakhil / "O"level/ equivalent examination	20%.
Total GPA of HSC/ Alim/ "A" level/ equivalent examination	20%.
Total	**100%**

The Students will be selected as per merit list prepared which will be given in the university website. Individual choice for selection of departments will be given preference as far as possible. In case of tie, merit position will be determined on the basis of marks obtained in admission test in

Mathematics, Physics, Chemistry and English respectively. Further dispute will be solved giving priority of result of HSC over SSC examination.

1.6.3 Medical Checkup

Candidates selected through above procedure will go for medical checkup in BAUST/CMH. If the medical authority considers any candidate unfit for study in university due to critical/contagious/mental diseases as shown in medical policy of university he/she will be declared unsuitable for admission.

CHAPTER - 2

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)

2.1 Introduction

Computer plays vital and in fact indispensable role in all fields of modern human activities. Consequently Computer Science and Engineering has established itself as one of' the most important branches of engineering. Recent development in computer has a considerable impact on society. It has already expanded to all fields of study starting from genetic engineering to space technology. Recent development in Artificial Intelligence has taken the human history a long way. That day is not very far when man can make machine like him.The Department of Computer Science and Engineering is one of the pioneer Departments of this Institute providing top-quality education in Computer Science and Engineering (CSE) at its undergraduate program. ICT is the leading booming sector in present day. In Bangladesh has already been declared as a thrust sector. Keeping this in mind the department offers CSE course to produce computer specialist.

2.2 Laboratory Facilities of the Department

The department endeavors to provide its faculty members and students adequate laboratory and other facilities. Departmental undergraduate courses are laboratory intensive and these requirements will be catered by following laboratories:

a) Software Engineering Lab
b) Networking Lab
c) Digital Lab
d) System Engineering Lab
e) Programming and Algorithm Lab
f) Microprocessor and Microcontroller Lab
g) Robotics and Interfacing Labs
h) VLSI and Automation Lab
i) Database and Data Warehouse Lab
j) Signal and Image Processing Lab
k) Operating System Lab (LINUX Based)

Students in Level 1 and Level 2 have to undertake laboratory classes in Physics, Chemistry, English, Electronics and Mechanical Engineering too.

Note: The laboratories of CSE Department are also used by the students of other departments for sessional classes and research work of relevant subject/courses if necessary. Similarly, if necessary undergraduate students of this department can access to the facilities of other departments and centers during project, thesis and research works.

CHAPTER - 3

RULES AND REGULATIONS FOR UNDERGRADUATE PROGRAMME

3.1 Number of Terms in a Year (Level)

From Level -1 to Level-4 there will be two regular terms (Tern-1 and Term-2) in each Level. In each term there will be compulsory 5/6 subjects and students have to pass at least 3/4 subjects for promotion to the next higher term. A student may appear in clearance examination for the subjects, if he/she fails in the term final examinations. From Level 1 Term 2 and above a student may have maximum of 4 back log subjects. Back log subjects are those subjects, which a student fails to pass evenafter clearance examination.If a student fails in any of the sessional in a term, he/she will not be promoted to next higher term even if he/she passes in all the theoretical subjects. A student may appear in the improvement examination for the subjects with letter grade less than 'B' in final examinations. The maximum letter grade in improvement examination will be 'B' and his/her former letter grade will be cancelled. A student will not be promoted to level 3 Term 1 until he/she clears all the Back Log subjects. The maximum letter grade in clearance and Back Log subjects will be 'C'.

3.2 Duration of Terms.

The duration of each of term will be as follows:

Ser	Events	Durations
1.	Classes before Mid Term Break	7 weeks
2.	Mid Term Break	1 weeks
3.	Classes after Mid Term Break	7 weeks
4.	Preparatory leave	2 weeks
5.	Term Final Examination	3 weeks
6.	Result	2 weeks
7.	Clearance/ Improvement/ Back Log Examination	2 weeks
	Total	**24 weeks**

3.3 Course Pattern and Credit Structure

The undergraduate program is covered by a set of theoretical courses along with a set of laboratory courses to support them.

3.3.1 Course Designation System

Each course is designated by a two to four letter code identifying the department offering the course followed by a four-digit number having the following interpretation:

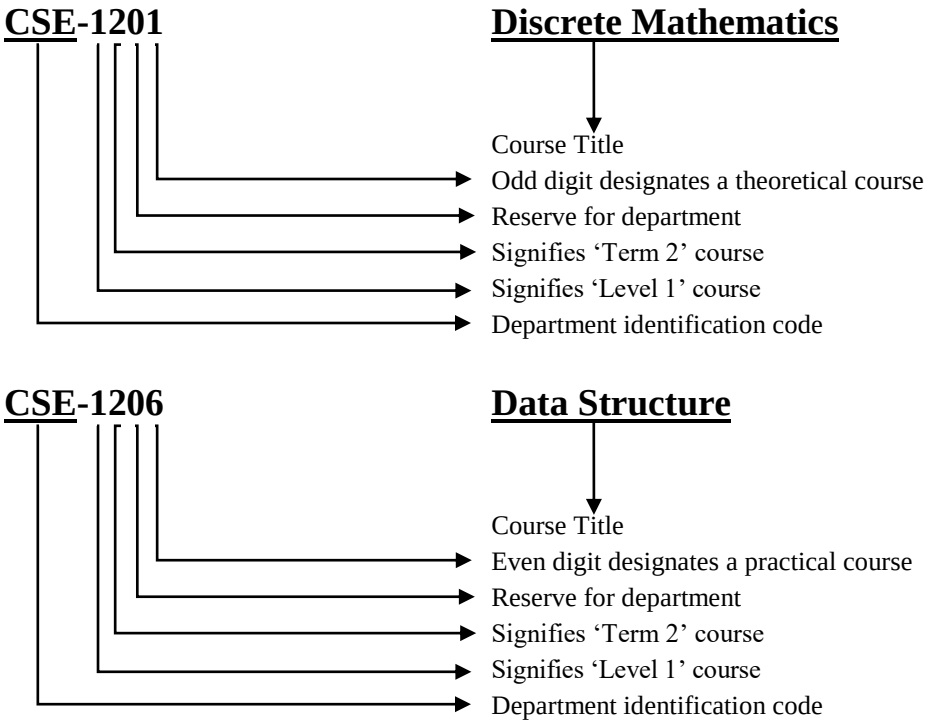
3.3.1.1 The first digit corresponds to the year/level in which the course is normally taken by the students.

3.3.1.2 The second digit corresponds to the term of that level.

3.3.1.3 The third digit is reserved for departmental use. It usually identifies a specific area/group of study within the department.

3.3.1.4 The last digit is an odd number for theoretical courses and an even number for sessional courses.

The course designation system is illustrated as Follows:



3.3.2 Assignment of Credits.

The assignment of credits to theoretical course is different from that of laboratory course, which is stated as follows:

3.3.2.1 For theoretical courses one lecture per week per term is equivalent to one credit.

3.3.2.2 For laboratory courses two class hours per week per term is equivalent to one credit.

3.3.2.3 Credits are also assigned to project work taken by the students. The total creditassigned to project work is 6 (3 for Level 4 Term 1 and 3 for Level 4 Term 2)

3.3.3 Types of Courses

The courses included in the undergraduate curricula are divided into the following groups:

Core Courses

In each discipline, a number of courses are identified as core courses, which form the nucleus of the respective bachelor degree program. A student has to complete the entire designated core courses of his/her discipline.

Prerequisite Courses

Some of the core courses are identified as prerequisite courses for a specific subject. A prerequisite course is one, which is required to be completed before some other course(s) can be taken.

Elective Courses

Apart from the core courses, the student can choose from a set of elective courses. A required number of optional courses from a specified group have to be chosen.

3.4 The Grading System

3.4.1 The Letter Grade

The total performance of a student in a given course is based on a scheme of continuous assessment. For theory courses this continuous assessment is made through a set of quizzes, class evaluation, class participation, homework, assignment and a term final examination. The assessment in laboratory courses is made by evaluating performance of the student at work during the class, viva-voce during laboratory hours and quizzes. Each course has a certain number of credits, which describes its corresponding weightages. A letter grade with a specified number of grade points is awarded in each course for which a student is registered. A student's performance is measured by the number of credits completed satisfactorily and by the weighted average of the grade points earned. A minimum grade point average (GPA) is essential for satisfactory progress. 160 credits have to be acquired in order to qualify for the degree. Letter grades and corresponding grade points will be awarded in accordance to the provisions shown below:

Numerical Markings	Grade	Grade Points
80% and above	A+	4.00
75% to below 80%	A	3.75
70% to below 75%	A-	3.50
65% to below 70%	B+	3.25
60% to below 65%	B	3.00
55% to below 60%	B-	2.75
50% to below 55%	C+	2.50
45% to below 50%	C	2.25
40%to below 45%	D	2.00
below 40%	F*	0.00
Incomplete	I	-
Withdrawal	W	-
Project/ Thesis continuation	X	-

* Subject in which the student gets 'F' grades will not be counted towards credit hours requirements and for the calculation of Grade Point Average (GPA)

3.4.2 Distribution of Marks

For Theory Courses

Thirty percent (30%) of marks of a theoretical course shall be allotted for continuous assessment, i.e. quizzes, class evaluation and class participation. The rest of the marks will be allotted to final examination. Distribution of marks for a given course is as follows:

Category	Marks %
Class Participation/ Observation	5%
Class Attendance	5%
Homework assignment and quizzes/class tests	20%
Final Examination (3 hours)	70%
Total	**100%**

The number of quizzes of a course shall be n+1, where n is the number of credits of the course. Evaluation of performance in quizzes will be on the basis of the best n quizzes. The scheme of continuous assessment that a particular teacher wishes to follow for a course will be announced on the first day of classes.

For Sessional Courses

The marks for the sessional courses will be distributed according to the type of the sessional course.

3.4.3 Calculation of CGPA

Grade Point Average (GPA) is the weighted average of the grade points obtained of all the courses passed/completed by a student. For example, if a student passes/completes *n* courses in a term having credits of C_1, C_2, … , C_n and his grade points in these courses are G_1, G_2, … , G_n respectively then

$$GPA = \frac{\sum_{i=1}^{n} C_i * G_i}{\sum_{i=1}^{n} C_i}$$

The Cumulative Grade Point Average (CGPA) is the weighted average of the GPA obtained in all the terms passed/completed by a student. For example, if a student passes/ completes n terms having total credits of TC1, TC2, …, TCn and his GPA in these terms are GPA1, GPA2, , GPAn respectively then

$$CGPA = \frac{\sum_{i=1}^{n} TC_i * GPA_i}{\sum_{i=1}^{n} TC_i}$$

A Numerical Example

Suppose a student has completed eight courses in a term and obtained the following grades:

Course	Credits,C_i	Grade	Grade Points, G_i	Ci*G_i
CSE-1101	2.00	A-	3.50	7.00
CSE-1102	0.75	A+	4.00	3.00
EEE-1163	3.00	B-	2.75	8.25
EEE-1164	1.50	C+	2.50	3.75
ME-1181	3.00	D	2.00	6.00
MATH-1141	3.00	A-	3.50	10.50
PHY-1103	3.00	A+	4.00	12.00
PHY-1104	0.75	B	3.00	2.25
HUM-1101	3.00	A	3.75	11.25
Total	**20.00**			**64.00**

GPA = 64.00/20.00 = 3.20

Suppose a student has completed four terms and obtained the following GPA:

Level	Term	Credit Hours Earned, TC_I	GPA Earned, GPA_i	GPA_i*TC_i
1	1	20.00	3.20	64.00
1	2	20.50	3.93	80.565
2	1	19.75	3.96	78.210
2	2	20.25	4.00	81.000
Total		**80.50**		**303.775**

CGPA = 303.775/80.50 = 3.77

3.4.4 Minimum Earned Credit and CGPA Requirement for Obtaining Degree

Minimum credit hour requirements for the award of bachelor degree in engineering (B.Sc. Engineering) and other discipline will be decided as per existing rules. The minimum CGPA requirement for obtaining a Bachelor degree in Computer Science and engineering and other discipline is 2.20. A student is required to complete the whole course of 160 credit within 4 years (8 terms). For an unavoidable reason if a student fails to complete the course within stipulated time of 4 years he/she must completes all degree requirements within a maximum period of 6 academic years (12 terms). Failure to complete all degree requirements within the given time frame may disqualify a student from continuation of his/her study at the university.

3.5 Attendance

All students are expected to attend classes regularly. The university believes that attendance is necessary for effective learning. The first responsibility of a student is to attend classes regularly, and one is required to attend at least 60% of all classes held in any course. Otherwise he/she shall not be allowed to sit for final examination.

3.6 Teacher-Student Interaction

The academic system in this university encourages students to come in close contact with the teachers. For promotion of high level of teacher-student interaction, a course coordinator (CC) is assigned to each course. Students are free to discuss with CC about all academic matters. Students are also encouraged to meet other teachers any time for help and guidance for academic matters. Heads of the departments, Proctor, Advisor of Students‟ Welfare (ASW), and Dean may address the students at some intervals.

3.7 Conduct and Discipline

During their stay in BAUST all students are required to abide by the existing rules, regulations and code of conduct. Students are strictly forbidden to form or be members of student organization or political party, club, society etc., other than those set up by university authority in order to enhance students' physical, intellectual, moral and ethical development. Zero tolerance in regards of sexual abuse, harassment in any forms, drug abuse and addiction are strictly observed in the campus.

CHAPTER – 4

COURSE REQUIREMENTS FOR UNDERGRADUATE COMPUTER SCIENCE AND ENGINEERING STUDENTS

4.1 Introduction.

The list of courses offered to the undergraduate students of Computer Science and Engineering (CSE) are categorized into Core courses and Elective courses. Some of the core courses are offered by the Department of CSE and some by other departments.

4.2 Core Courses.

The students have to complete all the core courses listed below:

4.2.1 List of Core Courses – CSE department

Sl No	Course Number	Course Name	Credit Hour
1.	CSE-1101	Introduction to Computer System	2.00
2.	CSE-1102	Introduction to Computer System Sessional	0.75
3.	CSE-1200	Software Development Project-I	0.75
4.	CSE-1201	Discrete Mathematics	3.00
5.	CSE-1203	Structured Programming Language	3.00
6.	CSE-1204	Structured Programming Language Sessional	1.50
7.	CSE-2101	Digital Logic Design	3.00
8.	CSE-2102	Digital Logic Design Sessional	1.50
9.	CSE-2103	Data Structures	3.00
10.	CSE-2104	Data Structures Sessional	1.50
11.	CSE-2105	Object Oriented Programming	3.00
12.	CSE-2106	Object Oriented Programming Sessional	1.50
13.	CSE-2200	Software Development Project-II	0.75
14.	CSE-2201	Algorithms	3.00
15.	CSE-2202	Algorithms Sessional	1.50
16.	CSE-2203	Object Oriented Programming II	3.00
17.	CSE-2204	Object Oriented Programming II Sessional	1.50
18.	CSE-2215	Digital Electronics and Pulse Technique	3.00
19.	CSE-2216	Digital Electronics and Pulse Technique Sessional	0.75
20.	CSE-2207	Numerical Methods	3.00
21.	CSE-2208	Numerical Methods Sessional	0.75
22.	CSE-3101	Database Management Systems	3.00

23.	CSE-3102	Database Management Systems Sessional	1.50
24.	CSE-3103	Microprocessor and Assembly Language	3.00
25.	CSE-3104	Microprocessor and Assembly Language Sessional	1.50
26.	CSE-3105	Computer Architecture	3.00
27.	CSE-3107	Communication Engineering	3.00
28.	CSE-3109	Theory of Computation	2.00
29.	CSE-3112	Technical Writing and Presentation	1.00
30.	CSE-3200	Web Engineering Project	1.50
31.	CSE-3201	Compiler	3.00
32.	CSE-3202	Compiler Sessional	0.75
33.	CSE-3203	Operating System	3.00
34.	CSE-3204	Operating System Sessional	0.75
35.	CSE-3205	Computer Network	3.00
36.	CSE-3206	Computer Network Sessional	1.50
37.	CSE-3207	Mathematical Analysis for Computer Science	3.00
38.	CSE-4100	Project and Thesis	6.00
39.	CSE-4101	Artificial Intelligence	3.00
40.	CSE-4102	Artificial Intelligence Sessional	0.75
41.	CSE-4103	Software Engineering and Information System	3.00
42.	CSE-4104	Software Engineering and Information System Sessional	0.75
43.	CSE-41XX	Option-I	3.00
44.	CSE-41XX	Option-II	3.00
45.	CSE-41XX	Option-II Sessional	1.50
46.	CSE-4201	Computer Graphics	3.00
47.	CSE-4202	Computer Graphics Sessional	0.75
48.	CSE-4203	Digital Signal Processing	3.00
49.	CSE-4204	Digital Signal Processing Sessional	1.50
50.	CSE-42XX	Option-III	3.00
51.	CSE-42XX	Option-III Sessional	0.75
52.	CSE-42XX	Option-IV	3.00
53.	CSE-42XX	Option-IV Sessional	0.75
		Total Credit Hours	**113.00**

4.2.2 List of Core Courses –Arts &Science department

SN.	Course Number	Course Name	Credit Hour
1.	MATH-1141	Math-I (Differential and Integral Calculus)	3.00
2.	PHY-1103	Physics	3.00
3.	PHY-1104	Physics Sessional	0.75
4.	HUM-1101	English	3.00
5.	MATH-1243	Math-II (Ordinary and Partial Differential Equations and Coordinate Geometry)	3.00

6.	CHEM-1201	Chemistry	3.00
7.	MATH-2145	Math-III (Vector Analysis, matrices and Fourier Analysis)	3.00
8.	MATH-2247	Math-IV(Complex Variable and Laplace Transform)	3.00
9.	HUM-3317	Society Ethics and Technology	2.00
10.	HUM-4215	Engineering Economics and Accounting	3.00
		Total Credit Hours	**26.75**

4.2.3 List of Core Courses –EEE department

SN.	Course Number	Course Name	Credit Hour
1	EEE-1163	Basic Electrical Engineering	3.00
2	EEE-1164	Basic Electrical Engineering Sessional	1.50
3	EEE-1269	Electronic Devices and Circuits	3.00
4	EEE-1270	Electronic Devices and Circuits Sessional	1.50
5	EEE-2169	Electrical Drives and Instrumentation	3.00
6	EEE-2170	Electrical Drives and Instrumentation Sessional	0.75
		Total Credit Hours	**12.75**

4.2.4 Core course –ME department

SN.	Course Number	Course Name	Credit Hour
1	ME-1181	Basic Mechanical Engineering	3.00
2	ME-1250	Engineering Drawing Sessional	1.50
3	ME-4117	Industrial Management	3.00
		Total Credit Hours	**7.50**

Undergraduate students of the Department of Computer Science and Engineering (CSE) have to follow a particular course schedule, the term-wise distributions of which are given below:

LEVEL-1, TERM-1					
Course No.	**Course Title**	**Hours/Week**		**Credit**	**Prerequisite**
		Theory	**Sessional**		
CSE-1101	Introduction to Computer System	2.00	-	2.00	
CSE-1102	Introduction to Computer System Sessional	-	1.50	0.75	
EEE-1163	Basic Electrical Engineering	3.00	-	3.00	
EEE-1164	Basic Electrical Engineering Sessional	-	3.00	1.50	
ME-1181	Basic Mechanical Engineering	3.00	-	3.00	
MATH-1141	Math-I (Differential and Integral Calculus)	3.00	-	3.00	
PHY-1103	Physics	3.00	-	3.00	
PHY-1104	Physics Sessional	-	1.50	0.75	
HUM-1101	English	3.00	-	3.00	
	Total	**17.00**	**6.00**	**20.00**	

LEVEL-1, TERM-2					
Course No.	**Course Title**	**Hours/Week**		**Credit**	**Prerequisite**
		Theory	**Sessional**		
CSE-1200	Software Development Project-I	-	1.50	0.75	
CSE-1201	Discrete Mathematics	3.00	-	3.00	
CSE-1203	Structured Programming Language	3.00		3.00	CSE-1101
CSE-1204	Structured Programming Language Sessional	-	3.00	1.50	
EEE-1269	Electronic Devices and Circuits	3.00	-	3.00	EEE-1163
EEE-1270	Electronic Devices and Circuits Sessional	-	3.00	1.50	
ME-1250	Engineering Drawing Sessional	-	3.00	1.50	
MATH-1243	Math-II (Ordinary and Partial Differential Equations and Coordinate Geometry)	3.00	-	3.00	MATH-1141
CHEM-1201	Chemistry	3.00	-	3.00	
	Total	**15.00**	**10.50**	**20.25**	

LEVEL-2, TERM-1					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-2101	Digital Logic Design	3.00	-	3.00	
CSE-2102	Digital Logic Design Sessional	-	3.00	1.50	
CSE-2103	Data Structures	3.00	-	3.00	CSE-1203
CSE-2104	Data Structures Sessional	-	3.00	1.50	
CSE-2105	Object Oriented Programming I	3.00	-	3.00	CSE-1203
CSE-2106	Object Oriented Programming I Sessional	-	3.00	1.50	
EEE-2169	Electrical Drives and Instrumentation	3.00	-	3.00	EEE-1269
EEE-2170	Electrical Drives and Instrumentation Sessional	-	1.50	0.75	
MATH-2145	Math-III (Vector Analysis, Matrices and Fourier Analysis)	3.00	-	3.00	MATH-1243
	Total	**15.00**	**10.50**	**20.25**	

LEVEL-2, TERM-2					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-2200	Software Development Project-II		1.50	0.75	
CSE-2201	Algorithms	3.00	-	3.00	CSE-1101, CSE2103
CSE-2202	Algorithms Sessional	-	3.00	1.50	
CSE-2203	Object Oriented Programming II	3.00		3.00	CSE-2105
CSE-2204	Object Oriented Programming II Sessional		3.00	1.5	
CSE-2205	Digital Electronics and Pulse Technique	3.00	-	3.00	EEE-1269, CSE-2101
CSE-2206	Digital Electronics and Pulse Technique Sessional	-	1.50	0.75	
CSE-2207	Numerical Methods	3.00	-	3.00	
CSE-2207	Numerical Methods Sessional	-	1.50	0.75	
MATH-2247	Math-IV (Complex Variable and Laplace Transform)	3.00	-	3.00	MATH-2145
	Total	**15.00**	**10.50**	**20.25**	

LEVEL-3, TERM-1					
Course No.	**Course Title**	**Hours/Week**		**Credit**	**Prerequisite**
		Theory	**Sessional**		
CSE-3101	Database Management Systems	3.00	-	3.00	CSE-1201
CSE-3102	Database Management Systems Sessional	-	3.00	1.50	
CSE-3103	Microprocessor and Assembly Language	3.00	-	3.00	CSE-2101
CSE-3104	Microprocessor and Assembly Language Sessional	-	3.00	1.50	
CSE-3105	Computer Architecture	3.00	-	3.00	
CSE-3107	Communication Engineering	3.00	-	3.00	
CSE-3109	Theory of Computation	2.00		2.00	
CSE-3112	Technical Writing and Presentation	-	2.00	1.00	
HUM-3115	Society Ethics and Technology	2.00	-	2.00	
	Total	**16.00**	**8.00**	**20.00**	

LEVEL-3, TERM-2					
Course No.	**Course Title**	**Hours/Week**		**Credit**	**Prerequisite**
		Theory	**Sessional**		
CSE-3200	Web Engineering Project	-	3.00	1.50	
CSE-3201	Compiler	3.00	-	3.00	CSE-2217
CSE-3202	Compiler Sessional	-	1.50	0.75	
CSE-3203	Operating System	3.00	-	3.00	
CSE-3204	Operating System Sessional	-	1.50	0.75	
CSE-3205	Computer Network	3.00	-	3.00	
CSE-3206	Computer Network Sessional	-	3.00	1.50	
CSE-3207	Mathematical Analysis for Computer Science	3.00	-	3.00	
HUM-3215	Engineering Economics and Accounting	3.00	-	3.00	
	Total	**15.00**	**9.00**	**19.50**	

LEVEL-4, TERM-1					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-4100	Project and Thesis	-	6.00	3.00	
CSE-4101	Artificial Intelligence	3.00	-	3.00	CSE-1201, CSE-2202
CSE-4102	Artificial Intelligence Sessional	-	1.50	0.75	
CSE-4103	Software Engineering and Information System	3.00	-	3.00	
CSE-4104	Software Engineering and Information System Sessional	-	1.50	0.75	
CSE-41XX	Option-I	3.00	-	3.00	
CSE-41XX	Option-II	3.00	-	3.00	
CSE-41XX	Option-II Sessional	-	3.00	1.50	
ME-4117	Industrial Management	3.00	-	3.00	
	Total	**15.00**	**12.00**	**21.00**	

LEVEL-4, TERM-2					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-4100	Project and Thesis*	-	6.00	3.00	
CSE-4201	Computer Graphics	3.00	-	3.00	
CSE-4202	Computer Graphics Sessional	-	1.50	0.75	
CSE-4203	Digital Signal Processing	3.00	-	3.00	
CSE-4204	Digital Signal Processing Sessional	-	3.00	1.50	
CSE-42XX	Option III	3.00	-	3.00	
CSE-42XX	Option III Sessional	-	1.50	0.75	
CSE-42XX	Option-IV	3.00	-	3.00	
CSE-42XX	Option-IV Sessional	-	1.50	0.75	
	Total	**12.00**	**13.50**	**18.75**	

OPTION-I					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-4119	Advanced Algorithms	3.00	-	3.00	
CSE-4121	Basic Graph Theory	3.00	-	3.00	
CSE-4123	Fault Tolerant System	3.00	-	3.00	
CSE-4125	Basic Multimedia Theory	3.00	-	3.00	
CSE-4127	Data and Network Security	3.00	-	3.00	
CSE-4129	Object Oriented Software Engineering	3.00	-	3.00	
CSE-4131	Artificial Neural Networks and Fuzzy Systems	3.00	-	3.00	
CSE-4133	Distributed Algorithms	3.00	-	3.00	
CSE-4135	Bioinformatics	3.00	-	3.00	
CSE-4137	Robotics	3.00	-	3.00	
CSE-4139	Machine Learning	3.00	-	3.00	
CSE-4141	Applied Statistics and Queuing Theory	3.00	-	3.00	
CSE-4143	Computational Geometry	3.00	-	3.00	

OPTION-II					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-4145	Digital System Design	3.00	-	3.00	
CSE-4146	Digital System Design Sessional	-	1.50	0.75	
CSE-4147	Peripherals, Interfacing and Embedded Systems	3.00	-	3.00	
CSE-4148	Peripherals, Interfacing and Embedded Systems Sessional	-	1.50	0.75	
CSE-4149	Mobile Computing and Applications	3.00	-	3.00	
CSE-4150	Mobile Computing and Applications Sessional	-	1.50	0.75	
CSE-4151	Web Architecture	3.00	-	3.00	
CSE-4152	Web Architecture Sessional	-	1.50	0.75	
CSE-4153	Parallel Processing and Distributed System	3.00	-	3.00	
CSE-4154	Parallel Processing and Distributed System Sessional	-	1.50	0.75	

OPTION-III and IV					
Course No.	Course Title	Hours/Week		Credit	Prerequisite
		Theory	Sessional		
CSE-4241	VLSI Design	3.00	-	3.00	
CSE-4242	VLSI Design Sessional	-	3.00	1.50	
CSE-4243	Pattern Recognition	3.00	-	3.00	
CSE-4244	Pattern Recognition Sessional	-	3.00	1.50	
CSE-4245	Digital Image Processing	3.00	-	3.00	
CSE-4246	Digital Image Processing Sessional	-	3.00	1.50	
CSE-4247	Telecommunication Engineering	3.00	-	3.00	
CSE-4248	Telecommunication Engineering Sessional	-	3.00	1.50	
CSE-4249	Simulation and Modeling	3.00	-	3.00	
CSE-4250	Simulation and Modeling Sessional	-	3.00	1.50	
CSE-4251	Data Ware-housing and Data Mining	3.00	-	3.00	
CSE-4252	Data Ware-housing and Data Mining Sessional	-	3.00	1.50	
CSE-4253	Network Programming	3.00	-	3.00	
CSE-4254	Network Programming Sessional	-	3.00	1.50	
CSE-4255	Cryptography and Network Security	3.00	-	3.00	
CSE-4256	Cryptography and Network Security Sessional	-	3.00	1.50	

Summary

Summary of the credit hour requirement to get B.Sc. Engineering degree in Computer Science and Engineering (CSE) are as follows:

Level and Term	Hours/Week		Credits	No of Theory Courses
	Theory	Sessional		
Level-1, Term-1	17.00	6.00	20.00	6
Level-1, Term-2	15.00	10.50	20.25	5
Level-2, Term-1	15.00	10.50	20.25	5
Level-2, Term-2	15.00	10.50	20.25	5
Level-3, Term-1	16.00	8.00	20.00	6
Level-3, Term-2	15.00	9.00	19.50	5
Level-4, Term-1	15.00	12.00	21.00	5
Level-4, Term-2	12.00	13.50	18.75	4
Grand Total	**120.00**	**80.00**	**160.00**	**41**

CHAPTER-5

DETAIL OUTLINE OF UNDERGRADUATE COURSESOFFERED BY THE DEPARTMENT OFCOMPUTER SCIENCE AND ENGINEERING

LEVEL-1 TERM-1

CSE-1101 2.00 hours in a week, 2.00 Cr.
Introduction to Computer System

Introduction to computations; history of computing devices; Computers; Major components of a computer; Hardware: processor, memory, I/O devices; Software: Operating system, application software; Report writing and Presentation; Basic architecture of a computer; Basic Information Technology; Number system: binary, octal, hexadecimal, binary arithmetic; Basic programming concepts; Program development stages: flow charts; Programming constructs: data types, operators, expressions, statements; Introduction to Object Oriented Programming.

Reference Book(s):

1. Introduction to Computers – Peter Norton.
2. The Complete PC Upgrade and Maintenance Guide – Mark Minasi
3. Discovering Computer Fundamentals-Shelly/Vermaat
4. Teach Yourself C-Herbert Schidlt

CSE-1102 1.50 hours in a week, 0.75 Cr.
Introduction to Computer System Sessional

Laboratory works based on CSE-1101

EEE-1163 3.00 hours in a week, 3.00 Cr.
Basic Electrical Engineering

Fundamental electrical concepts and measuring units; Direct current (dc): Current, voltage, resistance, power and energy; Series/Parallel Circuits; Methods of network analysis and Network Theorems; Capacitors, Inductors and introduction to magnetic circuits. Alternating current (ac): Instantaneous current, voltage and power for various combinations of R, L and C circuits; Effective current and voltage, average power; Phasor representation of sinusoidal quantities; Sinusoidal Single-Phase Circuit Analysis; Introduction to three phase circuits; Power factor and power equation (Δ and Y circuits).

Reference Book(s):

1. Introductory Circuit Analysis - Robert L. Boylestad.
2. Alternating Current Circuits - Russel M Kerchner and George F Corcoran.
3. Fundamentals of Electric Circuits - Charles K Alexander, Mathew N O Sadiku.
4. Introduction to Electric Circuits - Richard C. Dorf, James A. Svoboda.

EEE-1164 3.00 hours in a week, 1.50 Cr.
Basic Electrical Engineering Sessional

Laboratory works based on EEE-1163

ME-1181 3.00 hours in a week, 3.00 Cr.
Basic Mechanical Engineering

Sources of energy: conventional and renewable; Introduction to IC engines, Refrigeration and Air conditioning systems; Statics of particles and rigid bodies; Forces in trusses and frames; Relative motion; Kinematics of particles: Newton's Second Law of Motion; Kinematics of rigid bodies; Introduction to Robotics; Plane, rotational and spatial motion with applications to manipulators; Geometric configurations: structural elements, linkage, arms and grippers; Motion characteristics.

Reference Book(s):

1. A Textbook Of Thermal Enginering - R S Khurmi
2. Introduction to Thermal Engineering- J. P. Vasandani
3. Refrigeration And Air Conditioning- AhmadulAmeen

MATH-1141 3.00 hours in a week, 3.00 Cr.
Math-I (Differential Calculus and Integral Calculus)

DIFFERENTIAL CALCULUS

Limit, continuity and differentiability, successive differentiation of various types of functions, Leibnit'z theorem, Rolle's theorem, Mean Value theorem, expansion in finite and infinite forms, Lagrange's form of remainder, Cauchy's form of remainder (expansion of remainder), expansions of functions differentiation and integration, indeterminate form, partial differentiation, Euler's theorem, tangent and normal, sub tangent and subnormal in Cartesian and polar coordinates, maxima and minima of functions of single variables, curvature, asymptotes.

INTEGRAL CALCULUS

Definition of integrations, integration by the method of substitution, integration by parts, standard integrals, integration by the method of successive reduction, definite integrals, definite integral properties and its use in summing series, Walli's formula, improper integrals, Beta function and Gamma function, multiple integral and its application, area, volume of solid of revolution, area under a plane curve in Cartesian and polar coordinates, area of the region enclosed by two curves in Cartesian and polar coordinate, arc lengths of curves in Cartesian and polar coordinates.

Reference Book(s):

1. A text Book of Differential Calculus – Rahman and Bhattachrjee.
2. Differential Calculus – Shanti Narayan.
3. Differential Calculus – Dr. B. D. Sharma.
4. Differential Calculus – Das and Mukhajee.
5. Integral Calculus – Rahman and Bhattacharjee.
6. Integral Calculus – Abu Eusuf.
7. Integral Calculus – Das and Mukhajee.

PHY-1103 — 3.00 hours in a week, 3.00 Cr.
Physics — Prereq.:Nill

Waves-Oscillations & Wave mechanics

Oscillations: Differential equation of simple harmonic oscillator, total energy and average energy, Combination of simple harmonic oscillations, spring-mass system, damped oscillation, forced oscillation, resonance, stationary wave, phase velocity, group velocity.

Wave mechanics: Fundamental postulates of wave mechanics, Schroding's equation (time dependent and time independent), Operators, Uncertainty principle, energy of a free particle.

Optics and Laser

Theories of light: Interference of light, Young's double slit experiment, Fresnel's bi-prism. Interference in thin films, Newton's rings, Interferometers, Diffraction of light: Fresnel and Fraunhofer diffractions, Diffraction by single slit, diffraction by double slits, diffraction gratings, Resolving power of optical instruments, Polarization of light: production and analysis of polarized light, polarization by double refraction, Brewster's law, Malus law, Nicole prism, , optical activity and polarimeter.

Laser: spontaneous and stimulated emission, Helium-Neon laser, laser applications, Fiber optics.

Structure of Matter and Electricity: Crystalline & non-crystalline solids, single crystal and polycrystalline solids, crystal system, co-ordination number, packing factor, Miller indices, defects in solids, Bragg's law, Bonds in solids, Introduction to energy band, distinction between metal, insulator and semiconductor. **Electricity**: Coulomb's law, electric field, Gauss' law and its application, electric potential, capacitors and capacitance, dielectrics on atomic view, dielectric and Gauss's law, Ohm's law, resistivity -an atomic view, current density and drift velocity, Ampere's law, Faraday's law; Lenz's law, self-inductance and mutual inductance.

Reference Book(s):

1. Waves and oscillation - Brijlal and Subramannyam
2. Concept of modern physics - Arthur Beiser
3. Modern Physics - B. L. Theraja
4. Concept of electricity and magnetism - Haq, Rafiqullah, Roy
5. Electricity and magnetism - K. K. Tewari
6. Physics part-I - Resnick and Halliday
7. Physics part-II - Resnick and Halliday
8. A Text Book of Optics - Brijlal and Subramannyam
9. B. Sc. Physics V-I - C.L. Arora
10. B. Sc. Physics V-II - C.L. Arora

PHY-1104 — 1.5 hours in a week, 0.75 Cr.
Physics Sessional

Laboratory works based on PHY-1103

Reference Book(s):

1. Practical Physics –Dr. Giasuddin and Shahabuddin.
2. Practical Physics –C.L Arora.

HUM-1101 3 hours in a week, 3.00 Cr.
English

English Phonetics, Vocabulary, English Grammar-determiners, modifiers, affixes, root word, head word, types of verbs, different types of Clauses, Sentence construction and different types of sentences, synthesis of sentences, Grammatical Correction; Comprehension, Business Communication, Quotation and Tenders, Job Letters. Paragraph writing, Precise writing, Amplification, Report writing, Situational writing- posters and advertisements, notice and memorandum, message writing, Communication Today; Short stories and Literary articles written by some prominent writers, Research Study, research methodology.

Reference Book(s):

1. Prose of Our Time – AhsanulHaque, Sirajul Islam Chawdhury and M. Shamsuddoha

LEVEL-1 TERM-2

CSE-1200 1.50 hours in a week, 0.75 Cr.
Software Development Project-I

Student will develop software with proper documentation as assigned by lab teachers based on courses of level 1.

CSE-1201 3.00 hours in a week, 3.00 Cr.
Discrete Mathematics

Mathematical Logic: propositional calculus and predicate calculus; Set theory**:** sets, relations, partial ordered sets, functions; Mathematical reasoning and proof techniques; Counting: permutations, combinations, principles of inclusion and exclusion; Discrete Probability; Recurrence relations and recursive algorithms; Growth of functions; Graph theory: graphs, paths, trees, cycles; Algebraic structures: rings and groups.

Reference Book(s):

1. Discrete Mathematics & Its Applications- Kenneth H Rosen

2. Discrete Mathematics with Applications (Elsevier, 2004) -Thomas Koshy.

CSE-1203 3.00 hours in a week, 3.00 Cr.
Structured Programming Language

Programming concepts; Program development stages; Flow charts; Structured programming language: data types, operators, expressions, control structures; Functions and program structure: Function basics, parameter passing conventions, scope rules and storage classes, recursion; Header files; Preprocessor; Pointers and arrays, Strings, Multidimensional array; User defined data types: structures, unions, enumerations; Input and Output: standard input and output, formatted input and output, file access; Variable length argument list; Pointer and it's uses; Command line parameters; Error Handling; Graphics; Linking; Library functions.

Reference Book(s):

1. Teach Yourself C - Herbert Schidlt
2. C: The Complete Reference - Herbert Schildt
3. C Programming Language – Dennis M. Ritche

CSE-1204 3.00 hours in a week, 1.50 Cr.
Structured Programming Language Sessional

Laboratory works based on CSE-1203

EEE-1269 3.00 hours in a week, 3.00 Cr.
Electronic Devices and Circuits

Introduction to semiconductors, p-n junction diode, I–V characteristics; Diode applications: half and full wave rectifiers, clipping and clamping circuits, regulated power supply; Bipolar Junction Transistor (BJT): principle of operation, Transistor circuit configurations (CE, CB, CC), BJT biasing, BJT Transistor modeling, small-signal analysis of single and multi- stage amplifiers, frequency response of BJT amplifier. Field Effect Transistors (FET): Principle of operation of JFET and MOSFET, Depletion and enhancement type MOSFETs, Switching circuits using FETs, CMOS, biasing of FETs, FET small signal analysis, Low and high frequency response of FETs; Operational amplifiers and its applications; Feedback and oscillators circuits; Operation, characteristics and application of SCR, TRIAC, DIAC and UJT; Introduction to IC fabrication processes.

Reference Book(s):
1. Electronic Devices and Circuit Theory -Robert L. Boylestad and Louis Nashelsky
2. Electronic Principles – Albert P. Malvino.

EEE-1270 3.00 hours in a week, 1.50 Cr.
Electronic Devices and Circuits Sessional

Laboratory works based on EEE-2169

ME-1250 3.00 hours in a week, 1.50 Cr.
Engineering Drawing Sessional

Introduction: Lettering, numbering and heading, instrument and their use, sectional views and isometric views of solid geometrical figures; Plan, elevation and section of multistoried buildings; Building services drawings, detailed drawing of lattice towers; Use of AutoCAD software.

Reference Book(s):
1.Metric Drafting – Paul Wallah,
2.Drafting Technology and Practice – William P. Spence,
3.Technical Drawing – Frederick E Giesecke, Alva Mitchell, Henry C. Spencer,
4. Mechanical Engineering Drawing-Dr. Amalesh Chandra Mandal and Dr. Md. Quamrul Islam

MATH-1243 3.00 hours in a week, 3.00 Cr.
Math-II (Ordinary and Partial Differential Equations andCoordinate Geometry)

ORDINARY DIFFERENTIAL EQUATIONS:
Formulation of Differential Equations; Solution of first order differential equations by various methods, Solution of differential equation of first order but higher degrees; Solution of general linear equations of second and higher orders with constant co-efficient, Solution of Eular's Homogeneous linear differential equations.

PARTIAL DIFFERENTIAL EQUATIONS:
Introduction, Linear and nonlinear first order differential equations; Standard forms; linear equations of higher order; Equation of second order with variable coefficients;

COORDINATE GEOMETRY:
Transformation of coordinates, axes and its uses; Equation of conies and its reduction to standard forms; Pair of straight lines; Homogeneous equations of second degree; Angle between the pair of straight lines; Pair of lines joining the origin lo the point of intersection of two given curves, circles; System of circles; Orthogonal circles: Radical axis, radical center, properties of radical axes; Coaxial circles and limiting points: Equations of parabola, ellipse and hyperbola in cartesian and polar coordinates; Tangents and formals; pair of tangents; Chord of contact; Chord In terms of Its middle points; Pole and polar parametric co-ordinates; Diameters; Conjugate diameters and then- properties; Director circles and asymptotes.

Reference Book(s):
1. Ordinary and Partial differential Equations – M. D. Raisenghania.
2. Differential Equations – M. L. Khanna.
3. Differential Equations – B. D. Sharma.
4. Differential Equations – P. N. Chatterjee.
5. A text book on of coordinate geometry with vector analysis-Rahman and Bhattachrjee

CHEM-1201 3.00 hours in a week, 3.00 Cr.
CHEMISTRY

Atomic structure, quantum numbers, electronic configuration, periodic table; Properties and uses of noble gases; Different types of chemical bonds and their properties: Molecular structure of compounds: Selective organic reactions; Different types of solutions and their compositions; Phase rule. Phase diagram of mono component system; Properties of dilute solution; Thermo chemistry, chemical kinetics, chemical equilibrium; ionization of water and pi concept; Electrical properties of Solution.

Reference Book(s):
1. General Chemistry- Darrell D. Ebbing and Steven D. Gammon
2. Chemistry of Engineering Material –MominulHaque.
3. Introduction to Modern Inorganic Chemistry- S. Z. Haider
4. Physical Chemistry- P. W. Atkins
5. Essential of Physical Chemistry- B.S. Bhal, Arun Bhal and G. D. Tuli

LEVEL-2 TERM-1

CSE-2101 3.00 hours in a week, 3.00 Cr.
Digital Logic Design

Number systems and codes; Digital logic: Boolean algebra, De-Morgan`s theorems, logic gates and their truth tables, canonical forms, combinational logic circuits, minimization techniques; Arithmetic and data handling logic circuits, decoders and encoders, multiplexers and de-mulitplexers; Flip-flops, race around problems; Counters; asynchronous counters, synchronous counters and their applications; Registers and basic memory unit; Synchronous and asynchronous logic design; Design of sequential circuit: State diagram; State minimizations and assignments; Pulse mode logic; Fundamental mode design; PLA design.

Reference Book(s):

1. Digital Logic and Computer Design-M. Morris Manno.
2. Digital Computer Electronics - Albert P. Malvino, Jerald A Brown

CSE-2102 3.00 hours in a week, 1.50 Cr.
Digital Logic Design Sessional

Laboratory works based on CSE-2101

CSE-2103 3.00 hours in a week, 3.00 Cr.
Data Structure

Internal data representation; Abstract data types; Algorithm performance and elementary asymptotic analysis (Introduction to Big-O notation); Elementary data structures: array, linked list, stack, queue, tree and tree traversal, graphs and graph representation , heap, binary search tree; Sorting algorithms; Searching: linear search and binary search; Advanced data Structures: balanced binary search trees, skip list, advanced heaps; Hashing.

Reference Book(s):

1. Data Structure – Seymour Lipschutz, Schem Series
2. Introduction to Algorithms – Thomas H Cormen
3. Data Structures - Reingold and Hansen
4. Data structures and algorithm – Hopcroft, Ullman
5. Data Structures and Algorithms in Java -Michael T. Goodrich, Roberto Tamassia
6. Classic Data Structures –DebasisSamanta

CSE-2104 3.00 hours in a week, 1.50 Cr.
Data Structure Sessional

Laboratory works based on CSE-2103

CSE-2105 3.00hours in a week, 4.00 Cr.
Object Oriented Programming I

Philosophy of Object Oriented Programming (OOP); Advantages of OOP over structured programming; Encapsulation, classes and objects, access specifiers, static and non-static members; Constructors, destructors and copy constructors; Array of objects, object pointers, and object references, In-line functions, friend functions, reference; Inheritance: single and multiple inheritance; Polymorphism: overloading, abstract classes, virtual functions and overriding; Exceptions; Object Oriented I/O, inserter, extractor; Template functions and classes; namespaces, overview of Standard Template Library; Multi-threaded Programming.

Reference Book(s):
1. Teach Yourself C++ - Herbert Schildt
2. Turbo C/C++ Complete Reference - Herbert Schildt
3. C++ – How to Program – Deitel&Deitel

CSE-2106 3.00 hours in a week, 1.50 Cr.
Object Oriented Programming I Sessional

Laboratory works based on CSE-2105 (C++)

EEE-2169 3.00 hours in a week, 3.00 Cr.
Electrical Drives and Instrumentation

Transformers: Transformation ratio equations, Losses, Ideal Transformer, Voltage regulation, Matching Transformer; Alternators: Faradays Law, Dynamo, Generated voltage equation, Voltage regulation, DC Generator; Synchronous motor and Induction motor; DC motor; Stepper motors; Thyristor and Microprocessor based speed control of motors. Instrumentation amplifiers: Differential, logarithmic and chopper amplifiers; Frequency and voltage measurements using digital techniques; Recorders and display devices; Spectrum analyzers and Logic analyzers; Data acquisition and Interfacing Io microprocessor based systems; Transducers: Types, principles and application of photovoltaic, piezoelectric, thermoelectric, variable reactance and opto-electronic transducers; Noise reduction in instrumentation.

Reference Book(s):
1. A Text Book of Electrical technology - B.L Theraja
2. Electrical Machinery and Transformers - Irving L. Kossow
3. A Course in Electrical and Electronic Measurements and Instrumentation - A.K. Sawhney
4. Electronic Instrumentation and Measurements - David A. Bell

EEE-2170 1.50 hours in a week, 0.75 Cr.
Electrical Drives and Instrumentation Sessional

Laboratory works based on EEE-2169

MATH-2145 3.00 hours in a week, 3.00 Cr.
Math-III (Vector Analysis, Matrices and Fourier Analysis)

VECTOR ANALYSIS

Scalars and vectors, equality of vectors; Addition and subtraction of vectors; Multiplication of vectors by scalars; Scalar and vector product of two vectors and their geometrical interpretation: Triple products and multiple products; Linear dependence and independence of vectors. Differentiation and integration of vectors together with elementary applications; Definition of line, surface and volume integrals; Gradient, divergence and curl of point functions, various formulae, Gauss's theorem, Stoke's theorem, Green's theorem.

MATRICES AND FOURIER ANALYSIS

Matrices: Definition of matrix; Algebra of matrices; multiplication of matrices, transpose of a matrix, inverse of matrix; rank and elementary transformations of matrices; Solution of linear equations; linear dependence and independence of vectors. quadratic forms, matrix polynomials, determination of characteristic root and vectors, null space and nullity of matrix, characteristic subspace of matrix.

Fourier Analysis: Real and complex form of Fourier series; Finite transform; Fourier Integral; Fourier transforms and their uses in solving boundary value problems of wave equations.

Reference Book(s):

1. Vector Analysis – Dr. Muhammad AbdusSattar.
2. Vector Analysis – M. D. Raisinghania.
3. Martices and Linear Transformations – Mohammad Iman Ali.
4. An Introduction to Matrices – S. C. Gupta.
5. Matrics – FrandAsyres, JR.

LEVEL-2 TERM-2

CSE-2200 1.50 hours in a week, 0.75 Cr.
Software Development Project-II

Students will work in groups or individually to develop high quality Software Project(s) including new l/O drivers or similar projects involving operating systems modules in different types of Data Base Systems or object oriented and visual languages, Students will write structured program and use properdocumentation.

CSE-2201 3.00 hours in a week, 3.00 Cr.
Algorithm

Introduction to algorithms; Correctness proof and techniques for analysis of algorithms; Master Theorem; Methods for the design of efficient algorithms: divide and conquer, greedy method, dynamic Programming; Graph algorithms: Basic search and traversal techniques, Topological sorting, Connected components, Spanning trees, Shortest paths, Flow algorithms; Lower bound theory; NPcompleteness, NP-hard and NP-complete problems; Coping with Hardness: backtracking, branch and bound, approximation algorithms.

Reference Book(s):

1. Algorithm Design, Wiley, 2002 - Michael T. Goodrich and Roberto Tamassia
2. Algorithms, McGraw-Hill, 2006 - SanjoyDasgupta, Christos H. Papadimitriou, UmeshVazirani.
3. Algorithm Design, Addison-Wesley, 2005 - Jon Kleinberg and ÉvaTardos.
4. Introduction to Algorithms – Thomas H Corman

CSE-2202 3.00 hours in a week, 1.50 Cr.
Algorithm Sessional

Laboratory works based on CSE-2201

CSE-2203 3.00 hours in a week, 3.00 Cr.
Object Oriented Programming II

Introduction to Java, Java Development Environment, JVM, Advances of JAVA over Computer Programming, Byte Codes, Variable, Operator and Data Types, Classes, Threads, Introduction to Applets, Swing and Servlets, Basic Error Handling, Concept of Threading, Execution of Java Programming in UNIX Operating System.

ULM diagram basic. The object-oriented approach within the context of software engineering, the language, basic (procedural) elements of language: what an Eiffel program is, what the instruction set is, and how to declare and use entities (variables) and routines; The concepts underlying the object-oriented approach: modularity, inheritance, and dynamic binding, case study from the management information-system domain; Environment matters: system configuration, interfacing with external software, and garbage collection.

Reference Book(s):

1. Java How to Program – Deitel and Deitel
2. Java: Complete Reference – H. Schildt
3. Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development – Craig Larman
4. Object-Oriented Software Engineering - Stephen Schach
5. Object-Oriented Software Engineering: Practical Software Development using UML and Java - Timothy Lethbridge, Robert Laganiere, Robert Laganiere

CSE-2204 1.50 hours in a week, 0.75 Cr.
Object Oriented Programming II Sessional

Laboratory works based on CSE-2204

CSE-2205 3.00 hours in a week, 3.00 Cr.
Digital Electronics and Pulse Technique

Diode Logic Gates, Transistor Switches, Transistor Gates, Open Collector and High Impedance Gates, MOS Gates; Digital Logic Families: TTL, ECL, IIL and CMOS Logic with Operation Details; Characteristics of Digital ICs: Propagation delay, Power dissipation, Figure of Merit, Fan out, and Noise immunity; Electronic Circuits for Flip Flops, Counters and Register, Memory Systems, PLAs; S/H circuits, A/D and D/A Converters with Applications; Linear Wave Shaping, Diode Wave Shaping Techniques, Comparator Circuits, Switching Circuits; Pulse Transformers, Pulse Transmission, Pulse Generation; Monostable, Bi-stable and AstableMultivibrator; Schmitt Trigger; Optically Coupled Oscillators; Blocking Oscillators and Time-base Circuit; Timing circuits; Simple voltage sweeps, linear current sweeps.

Reference Book(s):

1. Pulse, Digital and Switching waveforms- Jacob Millman and Herbert Taub.
2. Microelectronics: Digital and Analog Circuits and Systems- Jacob Millman.
3. Operational Amplifier and Linear Integrated Circuits- Robert Coughlin

CSE-2206 3.00 hours in a week, 1.50 Cr.
Digital Electronics and Pulse Technique Sessional

Laboratory works based on CSE-2205

CSE-2207 3.00 hours in a week, 3.00 Cr.
Numerical Methods

Introduction; Solution of algebraic and transcendental equations: method of iteration, False Position method, Newton-Rhapson method; Solution of simultaneous linear equations: Cramer's rule, iteration method, Gauss-Jordan Elimination method, Choleski's process; Interpolation: diagonal and horizontal difference, differences of a polynomial, Newton's formula for forward and backward interpolation, Spline interpolation; Integration: Gauss quadrature formula, Trapezoidal rule, Simpson's rule, Weddle's rule; Solution of ordinary differential equations: Euler's method, Picard's method, Milne's method, Taylor's

series method, Runge-Kutta method; Least squares approximation of functions: linear and polynomial regression, fitting exponential and trigonometric functions.

Reference Book(s):

1. Numerical Methods for Engineers – Steven C Chapra, Raymokand P Canale
2. Numerical Methods – E Balagurusamy
3. Numerical Methods -- S. Balachandra Rao, C.K.Shantha
4. Numerical Methods for Scientific and Engineering Computation -- M.K. Jain, Iyengar, R.K. Jain

CSE-2208 1.50 hours in a week, 0.75 Cr.
Numerical Methods Sessional

Laboratory works based on CSE-2207

MATH-2247 3.00 hours in a week, 3.00 Cr.
Math-IV (Complex Variable and Laplace Transform)

COMPLEX VARIABLE

Complex number system, General functions of a complex variable, Limits and continuity of a function of complex variable and related theorems, Complex function, differentiation and the Cauchy- Riemann Equations. Line integral of a complex function, Cauchy's Integral Formula, Liouville's Theorem, Taylor's and Laurent's Theorem, Singular Residues, Cauchy's Residue Theorem.

LAPLACE TRANSFORM

Definition. Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace transform. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Periodic function, some special theorems on Laplace transform, Partial fraction. Solutions of differential equations by Laplace transform. Evaluation of improper integral.

Reference Book(s):

1. Theory and Problems of Complex Variables – Marray R Sprigel.
2. Theory and functions of Complex Variables – Shanti Narayan.

LEVEL-3 TERM-1

CSE-3101 3.00 hours in a week, 3.00 Cr.
Database Management System

Introduction of database systems; Models: Entity-Relationship model, Relational model; Relational algebra; SQL; Advanced SQL; Some applications using SQL. Integrity constraint; Relational database design; File organization and retrieval, file indexing and hashing; Transaction manager; Concurrency controller; Recovery manager; Security system; Database administration; Introduction to advanced database management systems: distributed database, parallel database, data mining and warehousing, multimedia, object-oriented, object-relational, real-time database.

Reference Book(s):

1. Database System Concepts- Abraham Silbeschatz, Henry F. Korth, S. Sundarshan
2. Files and Databases- An Introduction, Peter D. Smith and G.M. Barnes, Addison-Wesley

CSE-3102 3.00 hours in a week, 1.50 Cr.
Database Management System Sessional

Laboratory works based on CSE-3101.

CSE-3103 3.00 hours in a week, 3.00 Cr.
Microprocessor and Assembly Language

Introduction to 8-bit, 16-bit, and 32-bit microprocessors: architecture,addressing modes, instruction set, interrupts, multi-tasking and virtualmemory; Memory interface; Bus interface; Arithmetic co-processor;Opcode, Processor status and Flag registers, Addressing modes, Branching and Looping, Interrupt Structures, I/O interfacing.
An overview of Intel 80X86 and Pentium Processors, RISC processors, CISC processor.

Assembly Language: Introduction to IMB PC Assembly Language, Assembly Language syntax, Program Data, Variables, Named constants, program structure, and memory models, Input/Output instruction, Running program, Program Segment Prefix. Status register, flow control, logic operation, data structure, arithmetic operation, array, string manipulation

Reference Book(s):

1. The Intel Microprocessors - Barry B Brey
2. Microprocessors and Interfacing - Douglas V. Hall
3. Microprocessors and Microcomputer- based system design -Mohamed Rafiquzzaman.

CSE-3104 3.00 hours in a week, 1.5 Cr.
Microprocessor and Assembly Language Sessional

Laboratory works based on CSE-3103.

CSE-3105 3.00 hours in a week, 3.00 Cr.
Computer Architecture

Fundamentals of computer Design; Processor Design; Datapaths Design : single cycle and multi cycle implementations; Control Unit design : hardware and micro programmed; Hazards; Exceptions; Fixed Point Arithmetic; Arithmetic Logic Unit (ALU) Design; System organization; Parallel Processing; Pipeline: pipelined datapath and control, superscalar and dynamic pipelining, Pipeline structure vector supercomputers; RISC Processor; Memory organization.

Reference Book(s):

1. Computer Organization and Design- David A Patterson
2. Computer Organization and Architecture- William Stalling

CSE-3107 3.00 hours in a week, 3.00 Cr.
Communication Engineering

Fundamentals: Communication Engineering Fundamentals, Waveforms Spectra, Periodic waveforms and its properties, Fourier series, Noise and its different types.

Amplitude Modulation: Amplitude modulation, Amplitude modulation index, Frequency spectrum for sinusoidal AM, AM broadcast Transmitter.

Frequency Modulation: Frequency Modulatio**n,** Sinusoidal FM, Frequency spectrum for Sinusoidal FM, FM transmitter. FM receiver, Phase Modulation.

Pulse Modulation, Pulse Codes Modulation (PCM), Quantization, Compression, PCM Receiver, Differential PCM, Delta Modulation, Sigma-Delta A/D conversion, Pulse Frequency Modulation (PFM), Pulse Time Modulation (PTM), Pulse Position Modulation (PPM).

Digital Communication: Digital Communication, Basic Digital Communication System, Synchronization, Asynchronous Transmission, Probability of Bit Error in Base band Transmission, Matched Filter, Eye Diagrams, Digital Carrier Systems, Amplitude Shift keying, Frequency Shift Keying, Phase Shift Keying, Carrier Recovery Circuits, Differential Phase Shift Keying, Error Control Coding, Block Control, Repetition Encoding, Parity Encoding, Convolution Encoding.

Propagation: Radio Wave Propagation, Mode of Propagation, Microwave Systems, Tropospheric Propagation, VHF/UHF Radio Systems.

Satellite Communication: Satellite Communication, Kepler's First and Second Law, Orbits, Geostationary Orbits, Power System.

Fiber Optic Communication: Fiber Optic Communication, Propagation within a Fiber, Modes of Propagation, Losses in Fibers, Light sources for Fiber optics, Photo detectors.

Reference Book(s):

1. Data Communication and Networking - Behrouz A Forouzan
2. Data and Computer Communication - William Stallings
3. Data Communication & Networks – R L Brewster

CSE-3109 2.00 hours in a week, 2.00 Cr.
Theory of Computation

Finite automata: deterministic finite automata, nondeterministic finite automata, equivalence and conversion of deterministic and nondeterministic finite automata, pushdown automata; Context free languages; Context free grammars; Pushdown automata; Regular languages: regular expressions, non-regular languages, the pumping lemma; Turing Machines: basic machines, configuration, computing with Turing machines, combining Turing machines; Un-decidability.

Reference Book(s):

1. Introduction to the Theory of Computation. CENGAGE Learning, 3rd ed., 2012 - M. Sipser.
2. Introduction to Automata Theory, Languages, and Computation. Addison-Wesley Longman Publishing Co., Inc., 3rd ed., 2006 - J. E. Hopcroft, R. Motwani, and J. D. Ullman.
3. Elements of the Theory of Computation. Upper Saddle River, NJ, USA: Prentice Hall PTR, 2nd ed., 1997 - H. R. Lewis and C. H. Papadimitriou.

CSE-3112 2.00 hours in a week, 1.00 Cr.
Technical Writing and Presentation

Issues of technical writing and effective oral presentation in Computer Science and Engineering; Writing styles of definitions, propositions, theorems and proofs; preparation of reports, research papers, these and books: abstract, preface, contents, bibliography and index; Writing of book reviews and referee reports; Writing tools: Latex; Diagram drawing software; presentation tools.

HUM-3115 2.00 hours in a week, 2.00 Cr.
Society Ethics and Technology

Historical Perspectives of Technology, Social Perspectives of Technology, Ethical Perspectives of Technology, Globalization and Human Rights, Information Technology, Biomedical Technology, Population and the Environment.

Reference Book(s):

1. Winston M., Edelbach R., Society, Ethics and Technology, Wadsworth Publishing, 4th Ed., 2008
2. Create or Perish: The Case for Inventions and Patents - Robert HRines
3. Engineering Ethics – Carles E Haris

LEVEL-3 TERM-2

CSE-3200 3.00 hours in a week, 1.50 Cr.
Web Engineering Project

Introduction: internet, history of the TCT/IP protocol, worldwide web; web servers: case of Apache, other web servers; webpage design: HTML, JavaScript; XML Schemas, their validation and transformation; dynamic WebPages with CGI, PHP or JSP and database access; Web services: SOAP, WSDL (Web Service Description Language), XML-RPC protocol; configuration, maintenance, monitoring and security.

Student will work in groups or individually to develop a web based application with proper documentation that they have to present within this semester.

Reference Book(s):
1. Learning Web App Development - Purewal, Semmy

CSE-3201 3.00 hours in a week, 3.00 Cr.
Compiler

Introduction to compiling; Basic issues; Lexical analysis and Scanning; Syntax analysis; Syntax-directed translation; Attribute Grammars and Semantic Analysis Semantic analysis; type-checking; issues with run-time environments – source language issues; Issues in the design of code generation, Intermediate code generation; Error management; Storage organization-storage allocation strategies, target machine run-time storage management; Code optimization: The principle sources of optimization, Peephole optimization, Optimization of basic blocks-Loops in flow graphs; Introduction to global data-flow analysis-Code improving transformations.

Reference Book(s):
1. Compilers Principles, Techniques and Tools -- Alfred V. Aho, Jeffrey D. Ullman

CSE-3202 1.50 hours in a week, 0.75 Cr.
Compiler Sessional

Laboratory works based on CSE-3201.

CSE-3203 3.00 hours in a week, 3.00 Cr.
Operating System

Introduction of Operating System, types of OS; Process: process managements, process states, job and process scheduling, CPU scheduling algorithms, process coordination, critical section problems, semaphores, Inter-Process Communication (IPC), classical IPC problems, multiprocessing and timesharing, message and mailbox etc.; Memory management: swapping, memory allocation schemes, Paging and segmentation, virtual memory, page replacement strategies, working sets, demand paging; Input/output: hardware/software, disk, disk scheduling algorithms, Secondary storage management, terminals, clocks; Deadlock: resource allocation, detection, prevention, avoidance and recovery; File

management; Operating system security; Main Features of Windows, UNIX, LINUX; Introduction to distributed operating systems.

Reference Book(s):
1. Modern Operating System- Andrew S. Tanenbaum
2. Operating Systems – William Stallings

CSE-3204 1.50 hours in a week, 0.75 Cr.
Operating System Sessional

Laboratory works based on CSE-3203

CSE-3205 3.00 hours in a week, 3.00 Cr.
Computer Networks

Introduction: Computer Networks and Applications, OSI reference model, TCP/IP model and terminology, Connectionless and Connection Oriented services, Service primitives, The ARPANET, Hubs, Bridges, and Switches, FDDI, Fast Ethernet; Routing Algorithm.

Physical Layer: Circuit switching and Packet switching, X-25 protocol, Frame relay and Cell relay, ATM reference model.

Medium Access Sub layer: Pure and slotted ALOHA, Persistent and Non persistent CSMA, CSMA with collision detection and collision free protocols, IEEE standard 802.3 and Ethernet.

Data Link Layer: Types of errors, framing, error detection & correction methods; Flow control, Stop & wait ARQ, Go-Back- N ARQ, Selective repeat ARQ, HDLC.

Network Layer: Internet address, classfull address, subnetting, static vs. dynamic routing, shortest path algorithm, flooding, distance vector routing, link state routing, ARP, RARP, IP, ICMP.

Transport Layer: UDP, TCP, Connection management, Addressing, Establishing and Releasing Connection, Congestion control algorithm, Flow control and Buffering, Multiplexing.

Presentation Layer: Data Compression techniques, Frequency Dependent Coding, Context Dependent Encoding.

Application Layer: Internet and intranets, Internet services and goals, DNS, SMTP, FTP, Telnet, HTTP, World Wide Web (WWW), DHCP and BOOTP.

Domain Name System: Name servers; Email and Its privacy; SNMP; HTTP; World Wide Web; Network security: Cryptography, DES, IDEA, public key algorithm; Authentication; Digital signatures, Principles of Reliable Data Transfer FTP.

Networking in Practice: Designing LAN, Cabling, Establishing Client- Server network, Configuring: Directory Server, Proxy server, FTP server, E-mail server, web server, DB server, Firewall, Network troubleshooting, network maintenance, network monitoring, Network programming.

Reference Book(s):
1. Computer Networks - Andrew S. Tanenbaum
2. Computer Networks: Protocols, Standards, and Interfaces - Uyless Black
3. Internetworking with TCP/IP: Principles, Protocols, Architecture - D. E. Comer
4. TCP/IP Illustrated Vol. I - W. R. Stevens
1. Complete Networking: A Top Down Approach Featuring the Internet – 3rd edition, James F. Kurose, Keith W. Ross

CSE-3206 3.00 hours in a week, 1.50 Cr.
Computer Networks Sessional

Laboratory works based on CSE-3105.

CSE-3207 3.00 hours in a week, 3.00 Cr.
Mathematical Analysis for Computer Science

Recurrent problems; Manipulation of sums; Number theory; Special numbers; generating functions; Probability Distributions and Expectations: total probability and Bayes' rule, discrete probability distributions, continuous probability distributions; Random variables; stochastic process; Markov chains (discrete parameter, continuous parameter, birth-death process); Queuing models (birth-death model, Monrovian model), open and closed queuing network; Application of queuing models.

Reference Book(s):
1. Concrete Mathematics - Knuth.
2. Introduction to Probability Models- Sheldon M. Ross.

HUM-3215 3 hours in a week, 3.00 Cr.
Engineering Economics and Accounting

Engineering Economics
Microeconomics: Definition of economics; Fundamentals of economics; Market and government in a modern economy; Basic elements of supply and demand; Choice and utility; indifference curve technique; Analysis of cost; Short run long run theory of production.

Macroeconomics: Key concept of macroeconomics; saving, consumption, investment; National income analysis; Inflation, Unemployment.

Development: Theories of developments; Banking system of Bangladesh, National Budget, Development partners(World Bank, Asian Development Bank, World Trade Organization, International Monetary Fund)

Accounting
Financial Accounting: Objectives and importance of accounting; Accounting as an information system. Computerized system applications in accounting. Recording system, double entry mechanism; account and their classification; Accounting equation: Accounting cycle: Journal, ledger, trial balance. Preparation of financial statements considering adjusting and closing entries; Accounting concepts (principles) and conventions. Financial statement analysis and interpretation: Ratio analysis. Cost and Management

Accounting: Cost concepts and classification; Overhead cost: meaning and classification; Distribution of overhead cost: Overhead recover method/rate; Job order costing: preparation of job cost sheet and question price, Inventory valuation: absorption costing and marginal/variable costing technique; Cost-Volume-Pro fit analysis: meaning, breakeven analysis, contribution margin analysis sensitivity analysis. Short-term investment decisions; relevant and differential cost analysis. Long-term investment decisions: capital budgeting, various techniques of evaluation of investments.

Reference Book(s):

1. Managerial Accounting – Garrison
2. Accounting Principle – Kieso
3. Economics by Samuelson
4. Economics by John Sloman
5. Economic Development by Michael Todaro
6. Money and Banking by Dudly g luckett

LEVEL-4 TERM-1

CSE-4100 6.00 hours in a week, 3.00 Cr.
Project or Thesis

Each student has to complete one Project or Thesis in the combined duration of two semesters of 4th year. In course CSE 4100 (Part-I), a student has to make a proposal defense. The defensed project has to be completed in the continuation course CSE-4100 (Part-I) and CSE 4200 (Part-II) in next semesters.

CSE-4101 3.00 hours in a week, 3.00 Cr.
Artificial Intelligence

Overview of AI, Knowledge representation, LISP and other Al programming languages; Review of Un-Informed Search Strategies and game playing; Informed search Strategies: A*, Heuristic functions, Memory Bounded Search (IDA*, SMA*); Iterative improvement Search (Hill Climbing, Simulated Annealing), constraint satisfaction problems. Review of Propositional logic, first order Logic, Introduction to Planning, Partial Order Planning. Bayesian Rule and its use in probabilistic reasoning; Belief Networks and Decision Networks; Learning Decision Trees; Learning General Logical descriptions-Hypothesis, Examples, Current Best Hypothesis Search, Least Commitment Search; Learning Neural and Belief Networks ANN, Perceptions, MFFN (Back propagation, Applications of Neural Networks, Bayesian Methods for learning Belief Networks, Generic Algorithm, Reinforced learning. Introduction to Natural Language Processing.

Reference Book(s):

2. Artificial Intelligence: a modern approach - Stuart Jonathan Russell, Peter Norvig
3. Artificial Intelligence: a new synthesis - Nils J. Nilsson

CSE-4102 1.50 hours in a week, 0.75 Cr.
Artificial Intelligence Sessional

Laboratory works based on CSE-4101.

CSE-4103 3.00 hours in a week, 3.00 Cr.
Software Engineering and Information System

Information System Design

Information and System; System Analysis and Systems Analyst; Information gathering techniques; Structured analysis of systems;

Feasibility Study: Concepts (abstraction, refinement, modularity and hierarchy) and classification, Introduction to modeling language (Use case diagram, Sequence diagram and Activity diagram), Cost benefit analysis; Project scheduling; System design techniques; User interface design

Software Engineering

Introduction to system engineering and software engineering; Software requirements analysis, modeling and specification;

Software Designing: principals, models, design patterns and specification

Software Testing: Objectives and principles, testability, testing design and implementation models and documentations, verification, validation and debugging; Quality factors and metrics for different software engineering phases; Software project management issues.

Reference Book(s):
1. Software Engineering A Practical Approach - Rogers Pressman
2. Software Engineering – Ian Somerville
3. Software Engineering Fundamental - Ali Behforooz& Fredrick J. Hudson

CSE-4104 1.50 hours in a week, 0.75 Cr.
Software Engineering and Information System Sessional

Laboratory works based on CSE-4103

CSE-41XX 3.00 hours in a week, 3.00 Cr.
Option-I

CSE-41XX 3.00 hours in a week, 3.00 Cr.
Option-II

CSE-41XX 3.00 hours in a week, 1.50 Cr.
Option-II Sessional

ME-4117 3.00 hours in a week, 3.00 Cr.
Industrial Management

Principles of Management; Financial Principles; Management of Innovation; Technology Strategy; Best Management Practices; Sales and Marketing; Ratio Analysis; Prelude Lobster; Designing and Building Yachts; Commoditization vs. Differentiation; Total Quality Management; Entrepreneurship; Strategic Planning and Management of Technology; Teradyne Business Plan; MIS: Introduction, Decision Support Systems, MIS in decision making, Development of communication skills.

Reference Book(s):
1. System Safety Engineering and Management - Harold E. Roland ,Brian Moriarty
2. Project Management, Planning and Control - Albert Lester
3. Management for Engineers, Scientists and Technologists - John V. Chelsom, Andrew C. Payne, Lawrence R. P. Reavill
4. Management Information Systems – A Managerial Perspective - Uma G. Gupta

LEVEL-4 TERM-2

CSE-4100 6.00 hours in a week, 3.00 Cr.
Project and Thesis

This course is a continuation of the course CSE 4100 (Part-I) from the previous semester. A student has to complete the defense project proposal, submit it by the end of the semester and make an oral defense of the project or thesis.

CSE-4211 3.00 hours in a week, 3.00 Cr.
Computer Graphics

Standard graphics primitives; Graphics hardware; Graphics pipeline; Coordinate convention; Scan conversion; Clipping; Modeling transformation; Viewing transformation; Projection transformation; Polygons and polygon meshes; Curves and surfaces; Hidden lines and surface removal; Fractals; Ray tracing; Light models; Color models; Graphics programming.

Reference Book(s):

1. Computer Graphics Principle and Practice -- James D Foley, Van Dam
2. Computer Graphics Using Open GL – F S Hill J R
3. OpenGL programming Guide-(Official guide to learning openly)

CSE4212 1.50 hours in a week, 0.75 Cr.
Computer Graphics Sessional

Laboratory works based on CSE-4211.

CSE-4203 3.00 hours in a week, 3.00 Cr.
Digital Signal Processing

Introduction to speech, image & data processing; Discrete time signals, sequences; Linear Constant Coefficient difference equation; Sampling continuous time signals; Two dimensional sequences and systems; Z-transform, Inverse Z-transform, H-transform; Frequency domain representation, discrete time systems and signals; Fourier series and Fourier Transform; Parseval's theorem; Equivalent noise definition of bandwidth; Convolution, Correlation and method of numerical integration; Computation of the DFT: Goertzel FFT, Chirp Z-transform algorithms.

Reference Book(s):

1. Digital Signal Processing - John G. Proakis&DimitrisManolakis
2. Discrete-Time Signal processing - Allan Oppenheim & Ronald Schafer
3. Digital Signal Processing-A practical approach - Emmanuel C. Ifeachor Barrie W. Jervis
4. Signals and Systems - Rodger Ziemer& William Tranter, D

CSE-4204 **1.50 hours in a week, 0.75 Cr.**
Digital Signal Processing Sessional

Laboratory works based on CSE-4203.

CSE-42XX **3.00 hours in a week, 3.00 Cr.**
Option III

CSE-42XX **1.50 hours in a week, 1.50 Cr.**
Option III Sessional

CSE-42XX **3.00 hours in a week, 3.00 Cr.**
Option III

CSE-42XX **1.50 hours in a week, 0.75 Cr.**
Option III Sessional

OPTION-I

CSE-4119 3 hours in a week, 3.00 Cr.
Advanced Algorithms

Randomized Algorithms: Las Vegas and Monte Carlo Algorithms; Randomized Data Structures: Skip Lists; Amortized Analysis: Different methods, Applications in Fibonacci Heaps; Lower Bounds: Decision Trees, Information Theoretic Lower Bounds, Adversary Arguments; Approximation Algorithms: Approximation Schemes, Hardness of Approximation; Fixed Parameter Tractability: Parameterized Complexity, Techniques of designing Fixed Parameter Algorithms, Examples; Online Algorithms: Competitive Analysis, Online Paging Problem, k-server Problem; External Memory Algorithms; Advanced Data Structures: Linear and Non-linear Methods

Reference Book(s):

1. An Introduction to Computational Learning Theory - Michael J. Kearns ,UmeshVazirani
2. Algorithm Design - Jon Kleinberg ,ÉvaTardos
3. An Introduction to Computational Learning Theory - Michael J. Kearns ,UmeshVazirani
4. Randomized Algorithms - Rajeev Motwani ,PrabhakarRaghavan
5. Probability and Computing: Randomized Algorithms and Probabilistic Analysis - Michael Mitzenmacher , Eli Upfal
6. Approximation Algorithms - Vijay V. Vazirani

CSE-4121 3 hours in a week, 3.00 Cr.
Basic Graph Theory

Graphs and their applications; Basic graph terminologies; Basic operations on graphs; Graph representations; Degree sequence and graphic sequence; Paths, cycles and connectivity; Euler tours; Hamiltonian cycles; Ear decomposition; Trees and counting of trees; Distance in graphs and trees; Graceful labeling; Matching and covering; Planar graphs; Digraphs; Graph coloring; Special classes of graphs.

Reference Book(s):

1. Introduction to graph theory - Douglas B West.
2. Introduction to Graph Theory, 4th Edition, Pearson Education Asia, 1995 - Robin J. Wilson

CSE-4123 3 hours in a week, 3.00 Cr.
Fault Tolerant System

Introduction of Fault Tolerant Systems and architectures; Goal and Application of Fault Tolerant computing, Fundamental Definitions, Design techniques to achieve fault Tolerance, Fault detection and location in combinational and sequential circuits; Fault test generation for combinational and sequential circuits; Digital simulation as a diagnostic tool; Automatic test pattern generator; Fault modeling; Automatic test equipment, Faults in memory, memory test pattern and reliability; Performance monitoring, self-checking circuits, burst error correction and triple modular redundancy; Maintenance processors.

Reference Book(s):
1. Design and Analysis of Fault Tolerant Digital System - Barry W. Johnson
2. Fault-Tolerant Systems - Israel Koren, C. Mani Krishna

CSE-4125 3 hours in a week, 3.00 Cr.
Basic Multimedia Theory

Multimedia system-introduction; Coding and compression standards; Architecture issue multimedia; Operating systems issues in multimedia - real-time OS issues, synchronization, interrupt handling; Database issues in multimedia – indexing and storing multimedia data, disk placement, disk scheduling, searching for multimedia document; Networking issues in multimedia - Quality-of service guarantees, resource reservation traffic specification, happing, and monitoring, admission control; Multicasting issues; Session directories; Protocols for controlling sessions; Security issues in multimedia -digital water – making partial encryption schemes for video streams; multimedia applications – audio and video conferencing, video on demand, voice over IP.

Reference Book(s):
1. Multimedia: Computing, Communications & Applications - Steinmetz R., Nahrstadt K

CSE-4127 3 hours in a week, 3.00 Cr.
Data and Network Security

Overview, Symmetric cipher, Classical encryption technique, Block cipher and the data encryption standard (DES), Triple DES, Introduction to finite fields, Advanced Encryption Standard, Contemporary Symmetric Ciphers, confidentiality using symmetric encryption public, Key encryption and Hash functions, Public-key Cryptography, RSA algorithm, Key management, Diffie-Hellman key exchange, Other Public Key Cryptosystem, Message Authentication and Hash function, Hash Algorithm, Digital Signatures and Authentication protocols, Network Security practice, Authentication application, Wireless Network Security, Electrical Mail security, IP security, Web security, System security, Intruders, Malicious software and Firewall, Legal and Ethical Aspects.

Reference Book(s):
1. Cryptography and Network Security – William Stallings
2. Cryptography and Network Security- Behrouz A. Forouzan

CSE-4129 3 hours in a week, 3.00 Cr.
Object Oriented Software Engineering

The object-oriented approach within the context of software engineering, the language, basic (procedural) elements of language: what an Eiffel program is, what the instruction set is, and how to declare and use entities (variables) and routines; The concepts underlying the object-oriented approach: modularity, inheritance, and dynamic binding, case study from the management information-system domain; Environment matters: system configuration, interfacing with external software, and garbage collection. Advanced issues involving exception handling, repeated inheritance, typing problems, and parallelism; object-oriented software engineering process, concentrating on specific guidelines facilitate the

translation OOAD to a maintainable Addresses verification and validation (V&V) issues of Eiffel software systems built in a software engineering context; Building reusable libraries; The building of a parallel linear algebra library (Paladin).

Reference Book(s):
1. Object-Oriented Software Engineering - Stephen Schach
2. Object Oriented Software Engineering: A Use Case Driven Approach - Ivar Jacobson
3. Object-Oriented Software Engineering: Practical Software Development using UML and Java - Timothy Lethbridge, Robert Laganiere, Robert Laganiere

CSE-4131 3 hours in a week, 3.00 Cr.
Artificial Neural Networks and Fuzzy Systems

Biological nervous system: the brain and neurons, Introduction to artificial neural network and fuzzy systems, Theory and application of Artificial neural networks and fuzzy logic; Multi-layer perception: Back propagation algorithm, Self-organization map, Radial basis network, Hop field network, Recurrent network, Fuzzy set theory, Failing Adaptive Linear (ADALINE) and Multiple Adaptive Linear (MADALINE) networks, Generating internal representation, Cascade correlation and counter propagation networks, Higher order and bi-directional associated memory, Lyapunov energy function, attraction basin, Probabilistic updates: simulated annealing, Boltzmann machine, Adaptive Resonance Theory (ART) network. ARTI. ART2. Fuzzy ART mapping (ARTMAF) networks. K.ohonen'8 feature .l\ Learning Vector Quantization (LVQ) networks, Logic control: Adaptive fuzzy neural network; Genetic algorithm and evolution compacting, Applications to control; Pattern recognition; Nonlinear system modeling, Speech and image processing.

Reference Book(s):
1. Introduction to Artificial Neural Systems, Jacek M. Zurada
2. Artificial neural systems: foundations, paradigms, applications, and implementations, Patrick K. Simpson

CSE-4133 3 hours in a week, 3.00 Cr.
Distributed Algorithm

Models of distributed computing, Synchrony, communication and failure concerns, Synchronous message-passing distributed systems, Algorithms in systems with no failures - Leader Election and Breadth-First Search algorithms, The atomic commit problem, Consensus problems - the Byzantine Generals Problem, Asynchronous message-passing distributed systems, Failure detectors, Logical time and vector clocks, Routing algorithms.

Reference Book(s):
1. Distributed Systems. S. Mullender (ed.). Addison-Wesley, 1993.
2. Introduction to Distributed Algorithms. G. Tel. Cambridge Univ. Press, 2000.

CSE-4135 3 hours in a week, 3.00 Cr.
Bioinformatics

Sequence similarity, homology, and alignment. Pairwise alignment: scoring model, dynamic programming algorithms, heuristic alignment, and pairwise alignment using Hidden Markov Models. Multiple alignment: scoring model, local alignment gapped and ungapped global alignment. Motif finding: motif models, finding occurrence of known sites, discovering new sites. Gene Finding: predicting reading frames, maximal dependence decomposition. Analysis of DNA microarray data using hierarchical clustering, model-based clustering, expectation-maximization clustering, Bayesian model selection.

Reference Book(s):
1. An Introduction to Bioinformatics Algorithm - Neil C. Jones, Pavel A. Pevzner

CSE-4137 3 hours in a week, 3.00 Cr.
Robotics

Introduction to robotics, overview of robot mechanisms, dynamics, and intelligent controls, planar and spatial kinematics, and motion planning; mechanism design for manipulators and mobile robots, multi-rigid-body dynamics, 3D graphic simulation; control design, actuators, and sensors; wireless networking, task modeling, human-machine interface, and embedded software mechanical design, rigid body velocity, Jacobean, inverse kinematics, redundant and parallel robots, trajectory control, face control and hap tics, Micro and Nan-robotics, mobile robots.

Reference Book(s):
1. Introduction to Robotics: Analysis, Control, Applications - Saeed B. Niku
2. Introduction to Robotics: Mechanics and Control - John J. Craig

CSE-4139 3 hours in a week, 3.00 Cr.
Machine Learning

Introduction to machine learning, Supervised, unsupervised and reinforcement learning, Unsupervised learning algorithms, Concept Learning, Decision Tree Learning, Attribute based and relational supervised learning algorithms, Artificial Neural network based learning algorithms, Bayesian Learning, Evaluating Hypothesis, Genetic algorithm and genetic programming, Reinforcement learning algorithms, Computational learning theory.

Reference Book(s):
1. Machine Learning, Tom Michael Mitchell
2. Introduction to Machine Learning, EthemAlpaydin

CSE-4141 3 hours in a week, 3.00 Cr.
Applied Statistics and Queuing Theory

Applied Statistics:
Introduction; Frequency distribution, Mean, median, Mode and other measure of central tendency

standard deviation and other measure of dispersion, Moments, Skewness and kurtosis, Elementary probability theory, Characteristics of distributions, elementary sampling theory, Estimation, Hypothesis testing and regression analysis. Probability distribution and expectations, discontinuous probability distribution, e.g. binomial, position and negative binomial. Continuous probability distributions, e.g. normal and exponential.

Queuing Theory:
Stochastic processes, Discrete time Markov chain and continuous time Markov Chain. Birth-death process in queuing. Queuing models: M/M/1,M/M/C,M/G/1,M/D/1,G/M/1 solution of network of queue-closed queuing models and approximate models. Application of queuing models in Computer Science.

Reference Book(s):
1. Applied Statistics - Rebecca (Becky) M. (Margaret) Warner
2. Applied Statistics for Engineers and Scientists - Jay L. Devore and Nicholas R. Farnum
3. An Introduction to Queuing Theory - U. Narayan Bhat
4. Probability, Markov Chains, Queues, and Simulation: The Mathematical Basis of Performance Modeling - William J. Stewart

CSE-4143 3 hours in a week, 3.00 Cr.
Computational Geometry

Drawing fundamental geometric objects: Basic concepts, algorithms and their complexity; Point inclusion problems, convexity testing; Polygon triangulations and polygon partitioning; Convex hulls in two-dimensional and three-dimensional spaces; Proximity analysis: Voronoi diagrams and Delaunay triangulations. Drawing Graphs: Styles and applications of graph drawing; Drawing of rooted trees and planar graphs.

OPTION-II

CSE-4145 3 hours in a week, 3.00 Cr.

Digital System Design

Digital system design Hierarchy; ASM charts; Hardware description language; Design using MSI and LSI components; Combinational and sequential circuit design with PLD's, Introduction to CPLD's & FPGA's; Design of memory subsystem using SRAM and DRAM; Design of various components of a computer: ALU, memory and control unit - hardwired and micro-programmed, Microprocessor based designs; Computer bus standards; Design using special purpose controllers. Introduction to Embedded Systems; Product design; Product development process; Modularity in Design Research Directions: Architecture innovations; Application domains; User interfaces Case
Studies;

Reference Book(s):

1. Digital Systems Design with FPGAs and CPLDs - Ian Grout
2. Digital Systems Design Using VHDL - Jr. Charles H. Roth, Lizy K. John
3. Digital Design and Computer Architecture - David Harris and Sarah Harris
4. High-Speed Digital System Design: A Handbook of Interconnect Theory and Design
Practices - Stephen H. Hall, , Garrett W. Hall, James A. McCall
5. Digital Computer Architecture – Malvino, Brown

CSE-4146 1.50 hours in a week, 0.75 Cr.

Digital System Design Sessional

Laboratory works based on CSE-3145.

CSE-4147 3 hours in a week, 3.00 Cr.

Peripherals, Interfacing and Embedded Systems

Interrupts, address space partitioning, A-to-D and D-to-A converters and some related chips. Interfacing ICs of I/O devices – I/O ports, Programmable peripheral interface, DMA controller, interrupt controller, communication interface, interval time, etc. IEEE 488 and other buses, interfacing with microcomputer. Interfacing I/O devices – floppy disk, hard disk, tape, CD-ROM & other optical memory, keyboard, mouse, monitor, plotter, scanner, etc. Microprocessor in Scientific Instruments and other applications – Display, Protective Relays, Measurements of Electrical quantities, Temperature monitoring system, water level indicator, motor speed controller, Traffic light controller, etc. Microprocessor based interface design

Introduction to Embedded system, The Embedded Design Life Cycle, Models of Computation, State Charts, General language Characteristics (SDL, Petri nets, Message Sequence Charts, UML, JAVA, HDL), Embedded System Hardware,(Input, Communication, Processing Unit, Memories, output) Embedded operating systems, middleware & Scheduling, Implementing, ASIC, Embedded Systems Hardware/Software design.

Reference Book(s):

1. Computer Peripherals, Author: KlilKinm
2. Embedded system design, Author: P. Marwedel
3. Embedded System Design: An Introduction to Processes, Tools and Techniques, Author: Arnold Berger, Arnold S. Berger

CSE-4148 1.50 hours in a week, 0.75 Cr.
Peripherals, Interfacing and Embedded Systems Sessional

Laboratory works based on CSE-4147.

CSE-4149 3 hours in a week, 3.00 Cr.
Mobile Computing and Applications

Basic concepts: Mobile computing; Mobile computing architecture, Mobile technologies, Anatomy of a mobile device, Applications of mobile computing, Technical issues for mobility, Mobile agents and process migration.

Introduction to Mobile Development Frameworks and Tools: Fully Centralized Frameworks and Tools, N-Tier Client–Server Frameworks and Tools, J2ME, WAP, Symbian EPOC, iPhone, Android, Windows CE.

Android application development: Getting started with android programming, Android architecture, Application framework and libraries, Android runtime, Linux kernel, Android user interface, Data persistence, Messaging and networking, Location Based Services, Developing android services, Android application publishing

The User Experience: The Small Screen Problem, The Unified Look and Feel Paradigm, The iPhone Human Interface Guidelines, The Blackberry User Interface Guidelines, Common User Interface Guidelines,

Security Issues in mobile computing: Security threats, Ensuring consistency and reliability.

The Future of Mobile Computing: Upcoming Technologies, Convergence of Media and Communication Devices.

Reference Book(s):

1. Mobile Computing Principles: Designing and Developing Mobile Applications with UML and XML, Cambridge University Press, 2004.
2. Mobile Computing: Theory and Practice: Author: GargKumkum
3. Beginning Android Application Development (Wrox Programmer to Programmer), Wrox Press, 2011.
4. Android Application Development: Programming with the Google SDK, O'Reilly, 2009.
5. App Inventor: Create Your Own Android Apps - David Wolber , Hal Abelson, Ellen Spertus, Liz Looney.
6. Wireless Java: Developing with J2ME, 2/e, Jonathan Knudsen, ISBN: 1-59059-077-5, Apress, 2003.
7. Programming Mobile Devices: An Introduction for Practitioners - TommiMikkonen, Wiley, 2007.

CSE-4150 1.50 hours in a week, 0.75 Cr.
Mobile Computing and Applications Sessional

Laboratory works based on CSE-4149.

CSE-4151 3 hours in a week, 3.00 Cr.
Web Architecture

The objective of this course is to introduce and explain the basic concepts of web architecture. Students of this course assume to have prior knowledge of computer network and programming languages as the prerequisite. A reasonable familiarity of java programming will be the added advantage. Throughout the course, the introductory concepts of web architectures for developing web applications will be studied. Students will learn how to write Java applications that share data across the Internet for games, collaboration, software updates, file transfer and more. A behind-the-scenes look at HTTP, CGI, Servlets, Enterprise Java Beans, ORM, which supports the Internet and the Web will be provided. This course explores the knowledge and the tools to create the next generation software that takes full advantage of the Internet.

Reference Book(s):
1. Great Web Architecture Author: Clay Andres
2. Web Application Architecture: Principles, Protocols and Practices Author: Leon Shklar

CSE-4152 1.50 hours in a week, 0.75 Cr.
Web Architecture Sessional

Laboratory works based on CSE-4151.

CSE-4153 3 hours in a week, 3.00 Cr.
Parallel Processing and Distributed System

Introduction: Trends towards parallel processing, Parallel processing mechanism, Multiprogramming and Time sharing, Parallel Computer Structures, Parallelism and Pipelining, Parallel processing applications, Speedup Performance Laws, Parallel Random Access Machines (PRAM) and VLSI model.
Hardware Technology: Advanced processor Technology, Superscalar and Vector processor, Shared memory organization, Design of Linear and Non linear Pipeline processor, Multiprocessor System Interconnects.

Pipelining and Vector Processing: Principles of Pipelining, Classification of pipelined processors, Instruction and Arithmetic pipeline design, Vector Processing principles, Vector processing requirements,

Designing Pipelined processors, Compound Vector processing, Recent Vector processors, Vectorization and Optimization methods.

Parallel Programming: Parallel Programming models, Parallel Languages and Compilers, Code Optimization and Scheduling, Loop Parallelization and Pipelining, Parallel Programming Environments, Shared-variable program structures, mapping programs onto Multicomputers.

Distributed System: Introduction, Distributed System Architectures, Communication in Distributed Systems, Distributed Middleware, Client/Server Design Issues, Inter-process communication, RPC , Distributed Objects and Remote Invocation , Virtualization & Code Migration, Naming, Distributed Synchronization & Coordination, Consistency & Replication in Distributed Systems, Fault Tolerance, Distributed Transactions , Security, P2P Systems, Cloud Computing, Grid Computing.

Distributed System Programming: Java RMI, CORBA, P2P, COM, DCOM, Multi Agent System, SOAP, Web Service.

Reference Book(s):

1. Advanced Computer Architecture Parallelism, Scalability, Programmability - Kai Hwang (Senior Consulting Edition)
2. Computer Architecture and Parallel Processing - Kai Hwang, Faye A. Briggs
3. Communication Architecture for distributed system - R. J. Cypser
4. Distributed Systems: Concepts and Design - George Coulouris, Jean Dollimore, and Tim Kindberg
5. Distributed Systems: Principles and Paradigms - Tanenbaum and Van Steen.

CSE-4154 1.50 hours in a week, 0.75 Cr.

Parallel Processing and Distributed System Sessional

Laboratory works based on CSE-4153.

OPTION-III

CSE-4241 3 hours in a week, 3.00 Cr.
VLSI Design

VLSI Design Methodology: Top-down Design Approach, Technology Trends and Design Automation Algorithms; Introduction to CMOS Inverters and Basic Gates; CMOS Fabrication Process and Layout; CMOS Circuit Characteristics and Performance Estimation; Buffer Circuit Design; Introduction Bi-CMOS Circuits; Complex CMOS Gates; CMOS Building Blocks - Adder, Comparator, Multiplier, Counter, and Shifter; Data Path and Memory structures.

Design Methodology and Tools; Hardware modeling - Hardware Modeling Languages, Logic Networks, State Diagrams, Data-flow and Sequencing Graphs, Behavioral Optimization; Floor Planning and Architecture Design; ASIC design using FPGA and PLDs.

Reference Book(s):

1. Modern VLSI Design – System-on-Chip Design - Wayne Wolf.
2. CMOS VLSI Design- A Circuit and System Perspective - Neil H.E. Weste, David Harris, and Ayan Banerjee.

CSE-4242 1.50 hours in a week, 0.75 Cr.
VLSI Design Sessional

Laboratory works based on CSE-4241

CSE-4243 3 hours in a week, 3.00 Cr.
Pattern Recognition

Pattern Recognition: introduction, importance, Statistical and Neural Pattern Recognition: Bayesian classifier, Bayes decision theory, discriminate functions and decision surfaces, Bayesian classifier for normal distribution, Linear classifiers: discriminate functions and decision hyper planes, perception algorithm, least squares methods; Kessler's construction, Nonlinear classifiers: two and three layer perceptions, back propagation algorithm; Template matching: optimal path searching techniques. Dynamic programming methods, Correlation methods; Context dependent classification: observable and hidden Markov models and Viterbi algorithm. Three problems of HMM and their application in Speech Recognition, Syntactic Pattern Recognition, Clustering algorithms.

Reference Book(s):

1. Pattern recognition – Sergios Theodoridis, Konstantinos Koutroumbas
2. Pattern Recognition - William Gibson

CSE-4244 3 hours in a week, 0.75 Cr.
Pattern Recognition Sessional

Laboratory works based on CSE-4243.

CSE-4145 3 hours in a week, 3.00 Cr.
Digital Image Processing

Digital image fundamentals: visual perception, Light and Electromagnetic Spectrum, Image Sensing and Acquisition, Image Sampling and Quantization, Some basic relationships between pixels, Linear and Nonlinear operations; image transforms: First Fourier Transform (FFT),Discrete Cosine Transform (DCT), Karhunen and Loeve Transform (KLT), Wavelet transform and sub-band decomposition; image enhancement in the frequency domain and image restoration techniques, image compression techniques, image compression standards: JPEG,MPEG, H.261, and H.263, Image Segmentation.

Reference Book(s):
1. Digital Image Processing. Second Edition. Rafael C. Gonzalez & Richard E.
2. Digital Image Processing (Implementation Using MATLAB) - Ashish Jain

CSE-4146 1.50 hours in a week, 0.75 Cr.
Digital Image Processing Sessional

Laboratory works based on CSE-4245

CSE-4247 3 hours in a week, 3.00 Cr.
Telecommunication Engineering

Overview of telecommunication: history, evolution, convergence of telecommunication and data networks; Transmission media: Characteristics and applications of twisted pairs, coaxial cables and optical fibers, Terrestrial and satellite microwave, radio waves, VSAT; Telephone operating principles: telephone equipment, description of the modern phone; Telephone switching systems: PSTN, PBX, Centrex, standards; Basics of communication systems: modulation, multiplexing; Switching system: circuit switching, packet switching; Voice over Internet Protocol (VoIP), Fax over IP network, voice over frame relay, and ATM, ACDs, call centers, computer integration; Data communication equipment: introduction to terminals, modems, RS-232 and other interfaces, modem types; Tele-Traffic analysis; Cellular telephony: Frequency reuse, frequency management, channel alignment, handoff strategies, FDMA, TDMA, CDMA and GSM, Introduction to satellite communication, Optical fiber communication, Submarine cables, Digital Radio Microwave, etc.

Reference Book(s):
1. Introduction to Telecommunication – Cole
2. Optical Fiber Communication Principles and Practice – John M Senior
3. Modern Digital and Analog Communication System – B P Lathi

CSE-4248 3 hours in a week, 0.75 Cr.
Telecommunication Engineering Sessional

Laboratory works based on CSE-4247.

CSE-4249 3 hours in a week, 3.00 Cr.
Simulation and Modeling

Simulation modeling basics: systems, models and simulation; Classification of simulation model; Steps in a simulation study; Concepts in discreet-event simulation: event scheduling vs. process interaction approaches, Time-advance mechanism, organization of a discreet-event simulation model; continuous simulation models; Combined discreet-continuous models; Monte Carlo simulation; Simulation of queuing systems. Building valid and credible simulation models: validation principles and techniques, statistical procedures (or comparing real-world observations and simulation outputs, input modeling; Generating random numbers and random variants; Output analysis. Simulation languages; Analysis and modeling of some practical systems , Random Number Generator, Random Variables, Probability Distribution.

Reference Book(s):

1. Simulation Modeling and Analysis - Law A. M., Kelton W. D.
2. Computer Aided Modeling & simulation - J. A. Spriet
3. Computer Simulation and Modeling - R. S. Lehman
4. System Simulation - G. Cordon

CSE-4250 3 hours in alternate week, 0.75 Cr.
Simulation and Modeling Sessional

Laboratory works based on CSE-4249.

CSE-4251 3 hours in a week, 3.00 Cr.
Data ware-housing and Data Mining

Introduction; Data warehousing and OLAP technology for data mining; Data preprocessing; Data mining primitives, languages and systems; Descriptive data mining: characterization and comparison; Association analysis; Classification and prediction; Cluster analysis; Mining complex types of data; Applications and trends in data mining.

Reference Book(s):

1. Data Mining and Data Warehousing - Bharat Bhushan Agarwal, Sumit Prakash Tayal
2. Data Warehousing, Data Mining, and OLAP - Alex Berson and Stephen J. Smith

CSE-4252 3 hours in alternate week, 0.75 Cr.
Data ware-housing and Data Mining Sessional

Laboratory works based on CSE-451.

CSE-4253 3 hours in a week, 3.00 Cr.
Network Programming

Basic Networking Software (Protocol stacks, TCP/IP, HTTP, etc) Internet architecture and history, Elementary socket programming in C, Low level networking, Ethernet, ARP, The network layer, IP, DHCP, NAT, The network layer, routing, IPv6, Transport layer protocols, TCP, UDP, The socket interface (writing clients and servers) Advanced socket programming, non-blocking sockets, Server design (forking, threads, preforking), daemons, Network Programming in Java, DNS, email, HTTP, cgi, cookies, P2P Web services (XML, JSP, SOAP, etc) XML, DTDs, Schemas, XML Parsing, XSLT, Client side scripting, Javascript, AJAX, Web server technologies, Tomcat, servlets, Web server technologies, JSP, Web server, technologies, RPCs, Java RMI, XML-RPC, CORBA, Server scripting languages, PHP, Ruby Web services, SOAP, WSDL, UDDI, The Semantic Web, RDF, OWL. Network security Cryptography, authentication, digital signatures, Network security, Kerberos, IPSec, SSL, Implementation of security, Anonymity on the Web, tor, Multimedia and VoIP, RTP

Reference Book(s):

1. UNIX Network Programming, Author: Richard Stevens, Volume 1, Second Edition, Sockets and XTI, Prentice Hall, 1998, ISBN 0-13-490012-X.
2.UNIX System Programming using C++, Author: Terrence Chan.
3. The Design of the Unix Operating System, Author: Maurice Bach, Prentice Hall. (Dated description of Unix internals, but very readable)

CSE-4254 3 hours in alternate week, 0.75 Cr.
Network Programming Sessional

Laboratory works based on CSE-453.

CSE-4255 3 hours in a week, 3.00 Cr.
Cryptography and Network Security

Cryptography

Overview: CryptographyOverview and Terminologies.

Symmetric Ciphers: Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Steganography, Simplified DES, Block Cipher Principles, The Data Encryption Standard, The Strength of DES, Block Cipher Design Principles, Evaluation Criteria for AES, The AES Cipher, Triple DES, Blowfish, RC5, Characteristics of Advanced Symmetric Block Ciphers, RC4 Stream Cipher, Placement of Encryption Function, Traffic Confidentiality, Key Distribution.

Number theory: Fields, algebraic closures, Integers - divisibility, primes, testing primes, factorization, Euclideanalgorithm.

Public-Key Encryption: Principles of Public-Key Cryptosystems, The RSA Algorithm, Key Management.

Network Security

Message Authentication: Authentication Requirements, Authentication Functions, Message Authentication Codes, Hash Functions, MD5 Message Digest Algorithm, Secure Hash Algorithm, Digital Signatures, Authentication Protocols.

Network Security Practice: Kerberos, Pretty Good Privacy, S/Mime, IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Web Security Considerations, Secure Socket Layer and Transport Layer Security.

System Security: Intruders, Intrusion Detection, Password Management, Viruses and Related Threats, Virus Countermeasures, Firewalls.

Reference Book(s):

1. Cryptography and Network Security Principles and Practice - W. Stallings
2. Cryptography and Network Security - BehrouzForouzan
3. Applied Cryptography - Bruce Schneier
4. Computer Security - Dieter Gollmann
5. Differential Crypt Analysis of the Data Encryption Standard - E. Biham and A. Shamir
6. Cryptography and Data Security - D. Denning
7. A Course on Number Theory and Cryptography - N. Kobliz

CSE-4256 3 hours in a week, 0.75 Cr.
Cryptography and Network Security Sessional

Laboratory works based on CSE-4255.